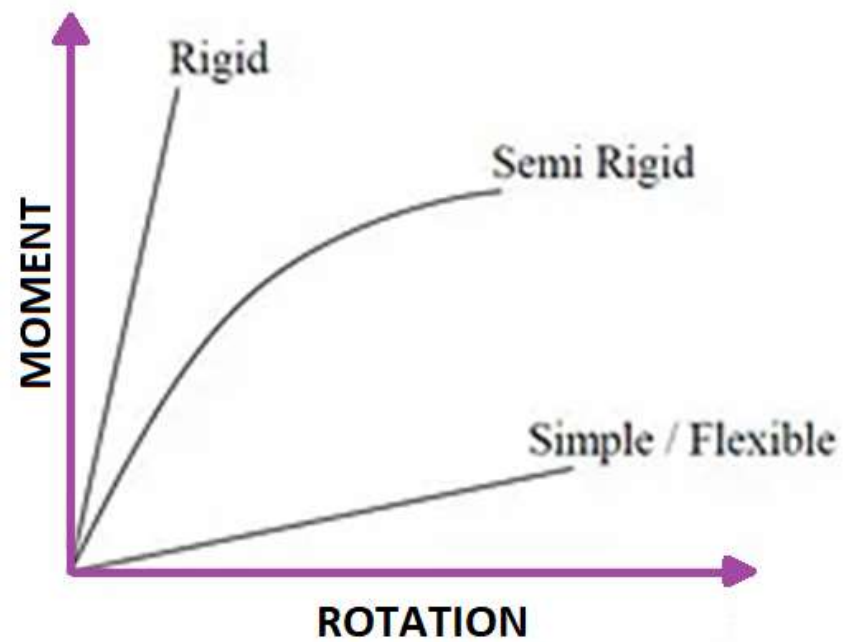
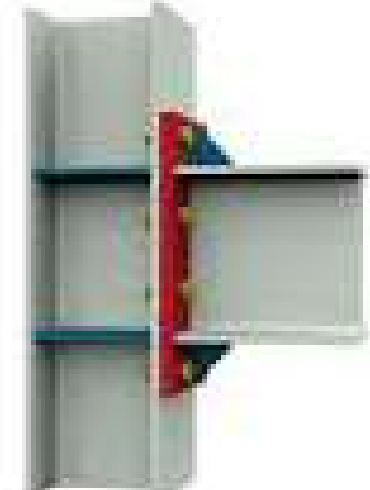
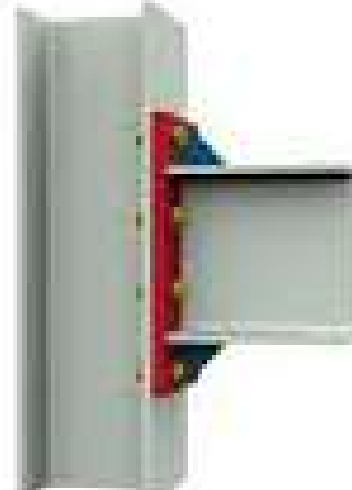
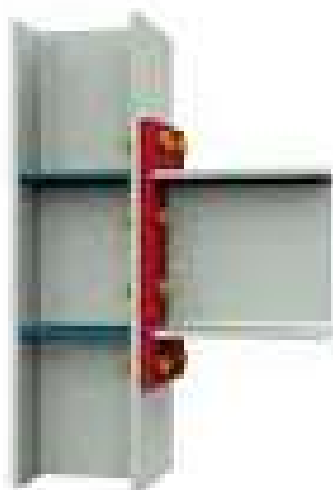
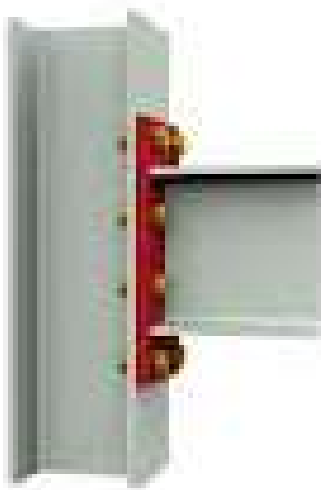
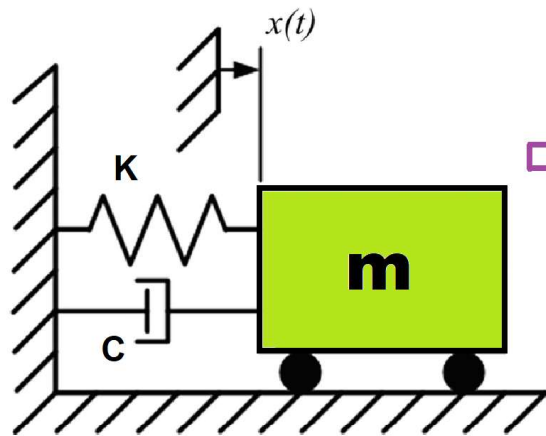


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

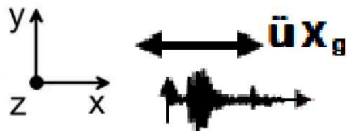
COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A SEMI-RIGID CONNECTION AS A SINGLE-DEGREE- OF-FREEDOM (SDOF) STRUCTURE: AN OPENSEES FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY ANALYSIS

THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)





➡ INCREMENTAL ROTATION



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

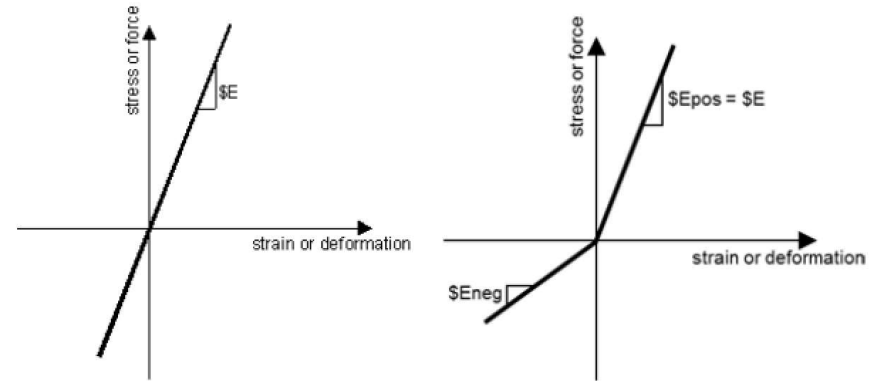
Δ_d = Lateral Displacement from Dynamic Analysis

Δ_y = Lateral Yield Displacement from Pushover Analysis

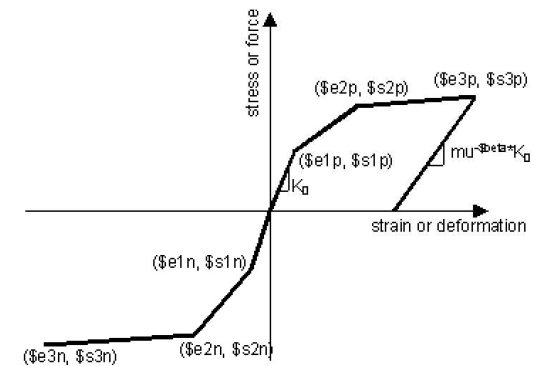
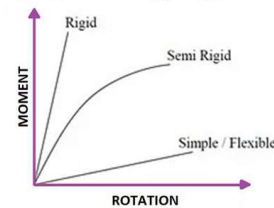
Δ_u = Lateral Ultimate Displacement from Pushover Analysis

$$P(t) = P_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$

$$m\ddot{u} + c\dot{u} + ku = P(t)$$

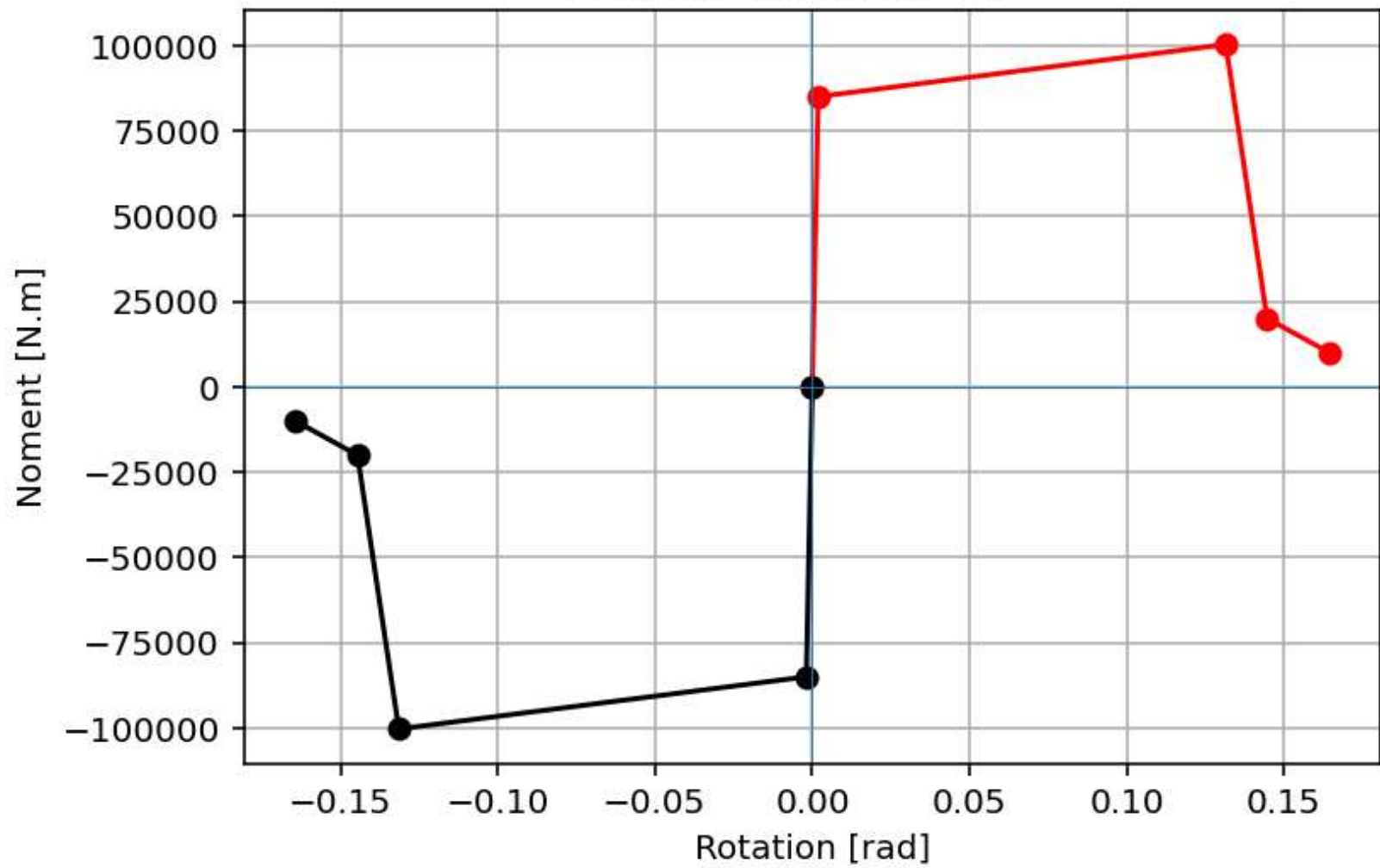


ELASTIC MATERIAL

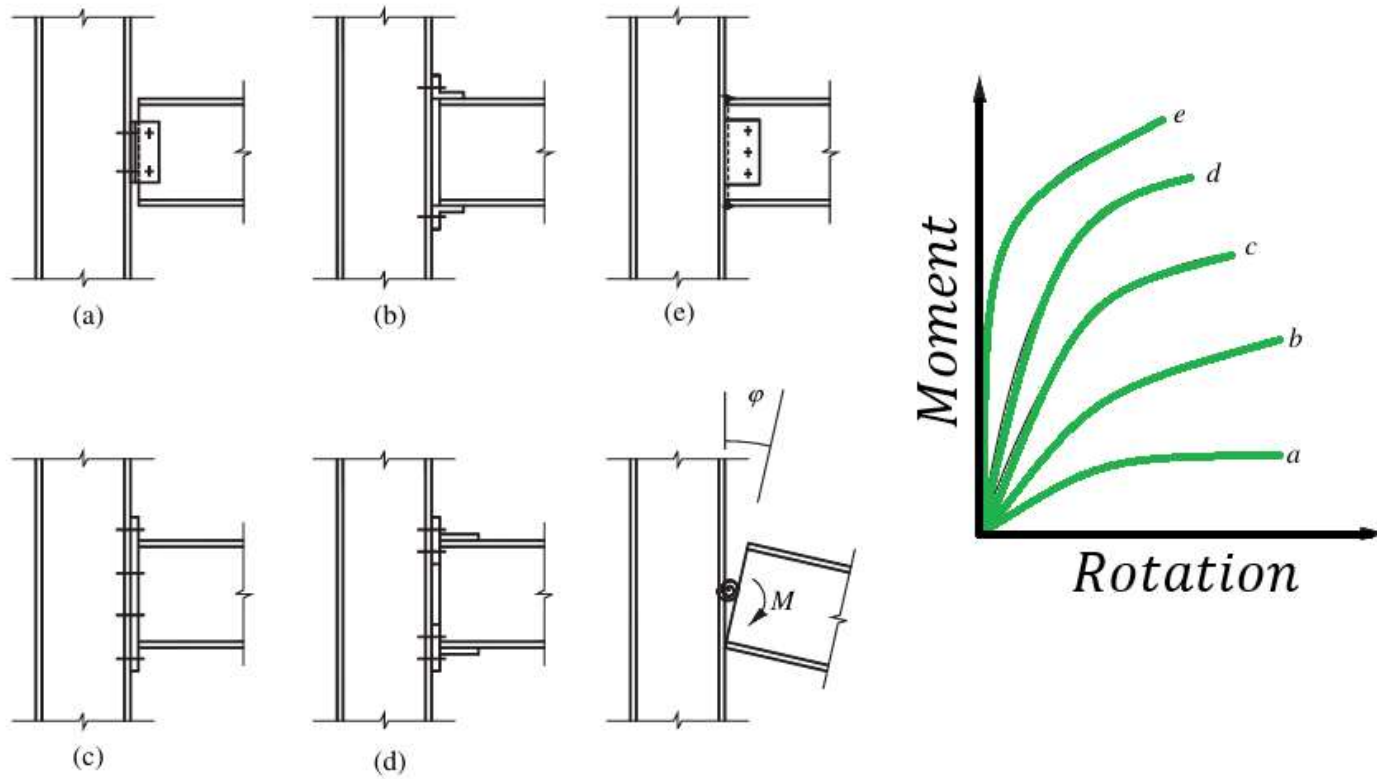


INELASTIC MATERIAL

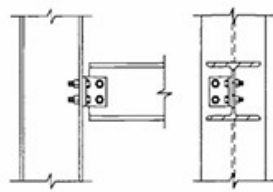
Moment-Rotation Curve



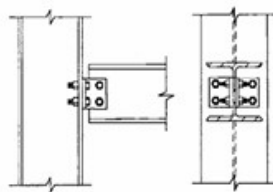
EFFECTS OF JOINT PANEL AND BEAM-COLUMN CONNECTION



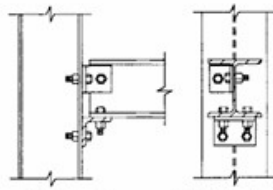
Connection configurations and the corresponding moment-rotation curves



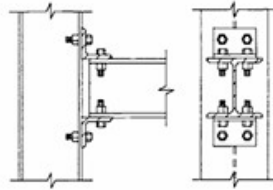
a. Single Web Angle



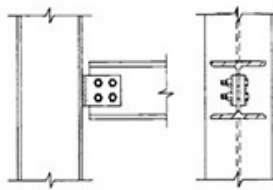
b. Double Web Angle



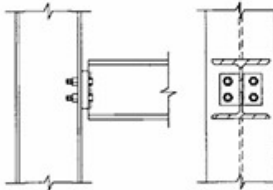
c. Bottom Flange and Web Angle



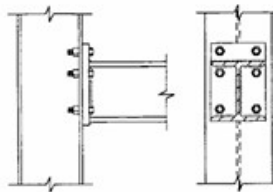
d. Top and Seat Angle



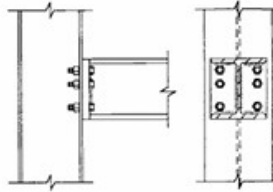
e. Web Side Plate



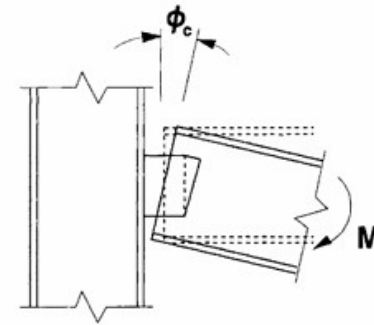
f. Header Plate



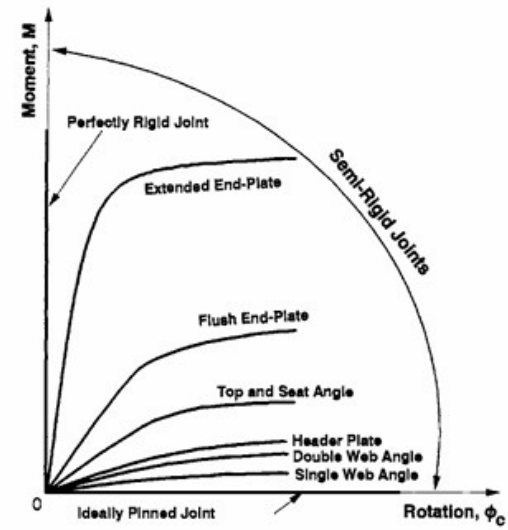
g. Extended End Plate



h. Flush End Plate



Rotational deformation of a connection.



Typical moment-rotation ($M-\phi_c$) curves of common connections.

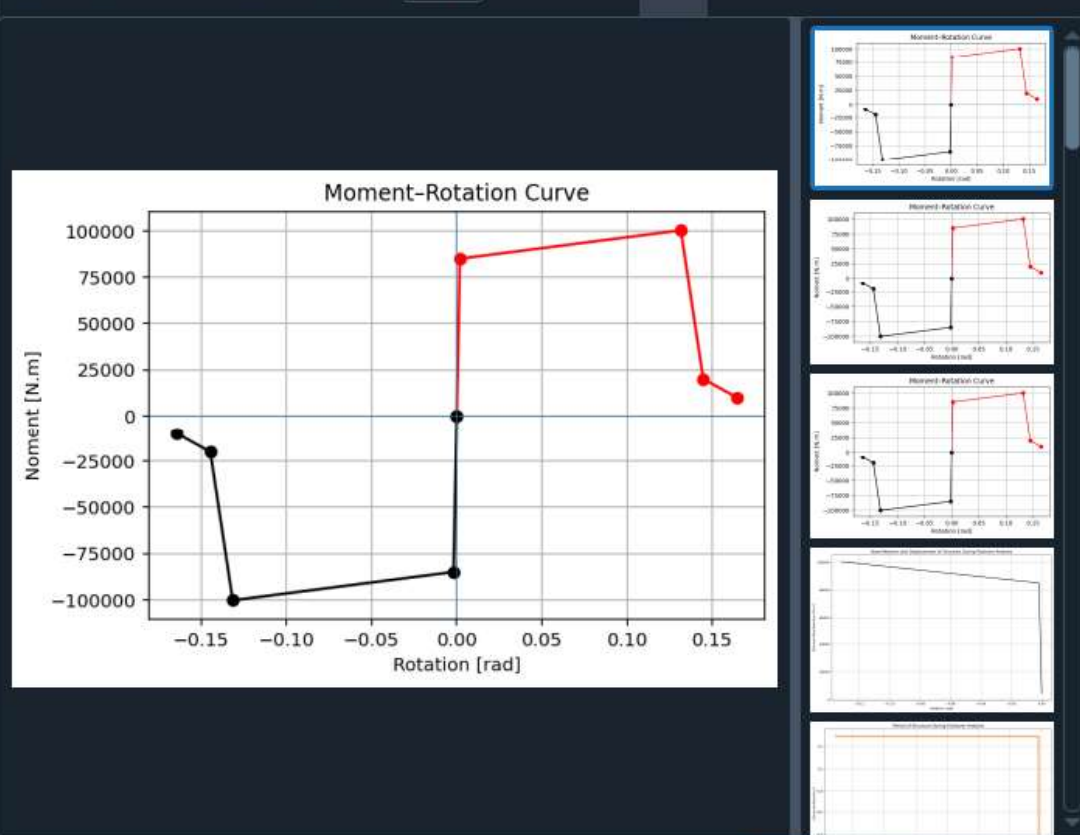
P(t)_SDOF_SEMI_RIGID_CONNECTION_8_ANA.py

```

191 DI = []
192
193 if ANAL_TYPE == 'PERIOD':
194     #PERIODmin, PERIODmax = S06.RAYLEIGH_DAMPING(1, 0.5*DR, DR, 0, 1)
195     PERIODmin, PERIODmax = S05.EIGENVALUE_ANALYSIS(1, PLOT=True)
196     # Compute modal properties
197     ops.modalProperties("-print", "-file", f"SALAR_ModalReport_{ANAL_TYPE}
198     return PERIODmin, PERIODmax
199
200 if ANAL_TYPE == 'STATIC':
201     ops.integrator('LoadControl', 1.0)
202     ops.analysis('Static')
203     OK = ops.analyze(1)
204     S02.ANALYSIS(OK, 1, MAX_TOLERANCE, MAX_ITERATIONS)
205     ops.reactions()
206     reaction.append(ops.nodeReaction(1, 1))
207     disp.append(ops.nodeDisp(center_node, 1))
208     DI.append(100*(np.abs(disp[-1])-DY)/(DSU-DY))
209     if DI[-1] <= 0.0: # DAMAGE INDEX -> 0.0 <= DI <=100
210         DI[-1] = 0.0
211     if DI[-1] >= 100:
212         DI[-1] = 100.0
213     print('\n\nSTATIC ANALYSIS DONE.\n\n')
214
215     DATA = (reaction, disp, DI)
216
217     return DATA
218
219 if ANAL_TYPE == 'PUSHOVER': # STATIC TIME-HISTORY ANALYSIS
220     IDctr1DOF = 1 # 1: Horizontal Displacment - 2: Vertical Dispalce
221     DINCR = -0.0001 # [m] Incremental Vertical Rotation
222     DMAX = -DSU # [rad] Max. Rotation
223     ops.integrator('DisplacementControl', center_node, IDctr1DOF, DINCR
224     ops.analysis('Static')

```

59 %



STATIC ANALYSIS

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...ON_8_ANA\P(t)_SDOF_SEMI_RIGID_CONNECTION_8_ANA.py

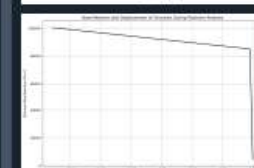
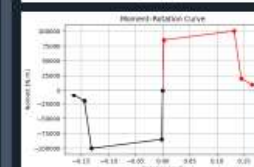
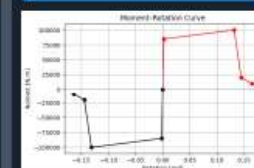
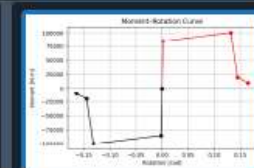
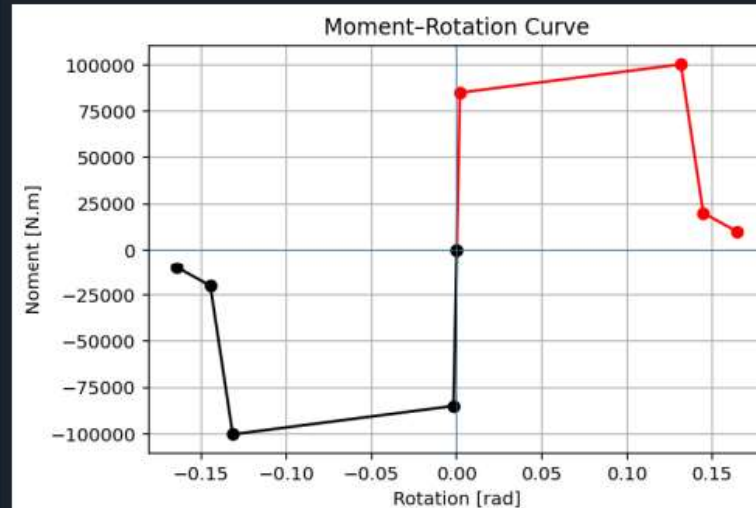
P(t)_SDOF_SEMI_RIGID_CONNECTION_8_ANA.py

```

191 DI = []
192
193 if ANAL_TYPE == 'PERIOD':
194     #PERIODmin, PERIODmax = S06.RAYLEIGH_DAMPING(1, 0.5*DR, DR, 0, 1)
195     PERIODmin, PERIODmax = S05.EIGENVALUE_ANALYSIS(1, PLOT=True)
196     # Compute modal properties
197     ops.modalProperties("-print", "-file", f"SALAR_ModalReport_{ANAL_TYPE}
198     return PERIODmin, PERIODmax
199
200 if ANAL_TYPE == 'STATIC':
201     ops.integrator('LoadControl', 1.0)
202     ops.analysis('Static')
203     OK = ops.analyze(1)
204     S02.ANALYSIS(OK, 1, MAX_TOLERANCE, MAX_ITERATIONS)
205     ops.reactions()
206     reaction.append(ops.nodeReaction(1, 1))
207     disp.append(ops.nodeDisp(center_node, 1))
208     DI.append(100*(np.abs(disp[-1])-DY)/(DSU-DY))
209     if DI[-1] <= 0.0: # DAMAGE INDEX -> 0.0 <= DI <=100
210         DI[-1] = 0.0
211     if DI[-1] >= 100:
212         DI[-1] = 100.0
213     print('\n\nSTATIC ANALYSIS DONE.\n\n')
214
215     DATA = (reaction, disp, DI)
216
217     return DATA
218
219 if ANAL_TYPE == 'PUSHOVER': # STATIC TIME-HISTORY ANALYSIS
220     IDctr1DOF = 1 # 1: Horizontal Displacment - 2: Vertical Dispalce
221     DINCR = -0.0001 # [m] Incremental Vertical Rotation
222     DMAX = -DSU # [rad] Max. Rotation
223     ops.integrator('DisplacementControl', center_node, IDctr1DOF, DINCR
224     ops.analysis('Static')

```

59 %



Python Console Files Help Variable Explorer Debugger Plots History

PUSHOVER ANALYSIS

File Edit Search Source Run Debug Consoles Projects Tools View Help

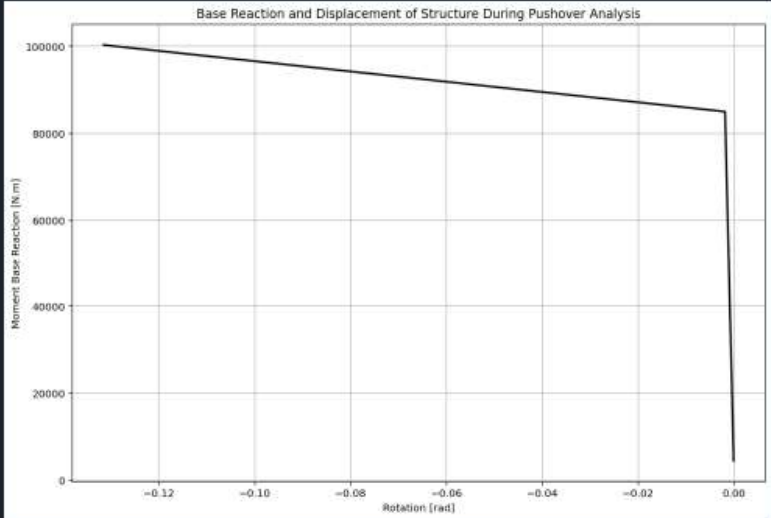
C:\Users\ DELL\Desktop\OPENSEES_FILES\+EIGHT_ANALY...ON_8_ANA\P(t)_SDOF_SEMI_RIGID_CONNECTION_8_ANA.py

P(t)_SDOF_SEMI_RIGID_CONNECTION_8_ANA.py X

```
218
219
220 if ANAL_TYPE == 'PUSHOVER': # STATIC TIME-HISTORY ANALYSIS
221     IDctrlDOF = 1 # 1: Horizontal Dispalcement - 2: Vertical Dispalce
222     DINCR = -0.0001 # [m] Incremental Vertical Rotation
223     DMAX = -DSU # [rad] Max. Rotation
224     ops.integrator('DisplacementControl', center_node, IDctrlDOF, DINCR
225     ops.analysis('Static')
226     Nsteps = int(np.abs(DMAX/ DINCR))
227     STEP = 0.0
228     for step in range(Nsteps):
229         OK = ops.analyze(1)
230         S02.ANALYSIS(OK, 1, MAX_TOLERANCE, MAX_ITERATIONS)
231         ops.reactions()
232         reaction.append(ops.nodeReaction(1, 1))
233         disp.append(ops.nodeDisp(center_node, 1))
234         # IN EACH STEP, STRUCTURE PERIOD GOING TO BE CALCULATED
235         #PERIODmin, PERIODmax = S06.RAYLEIGH_DAMPING(1, 0.5*DR, DR, 0,
236         PERIODmin, PERIODmax = S05.EIGENVALUE_ANALYSIS(1, PLOT=True)
237         PERIOD_MIN.append(PERIODmin)
238         PERIOD_MAX.append(PERIODmax)
239         DI.append(100*(np.abs(disp[-1])-DY)/(DSU-DY))
240         if DI[-1] <= 0.0: # DAMAGE INDEX -> 0.0 <= DI <=100
241             DI[-1] = 0.0
242         if DI[-1] >= 100:
243             DI[-1] = 100.0
244         STEP += 1
245         #print(STEP, disp[-1], reaction[-1])
246         print(f"Step: {STEP}, Rotation: {disp[-1]:.4f} rad, Moment-Reac
247     else:
248         print('\n\nPUSHOVER ANALYSIS DONE.\n\n')
249
250 DATA = (reaction, disp, DI,
251         np.array(PERIOD_MIN), np.array(PERIOD_MAX))
```

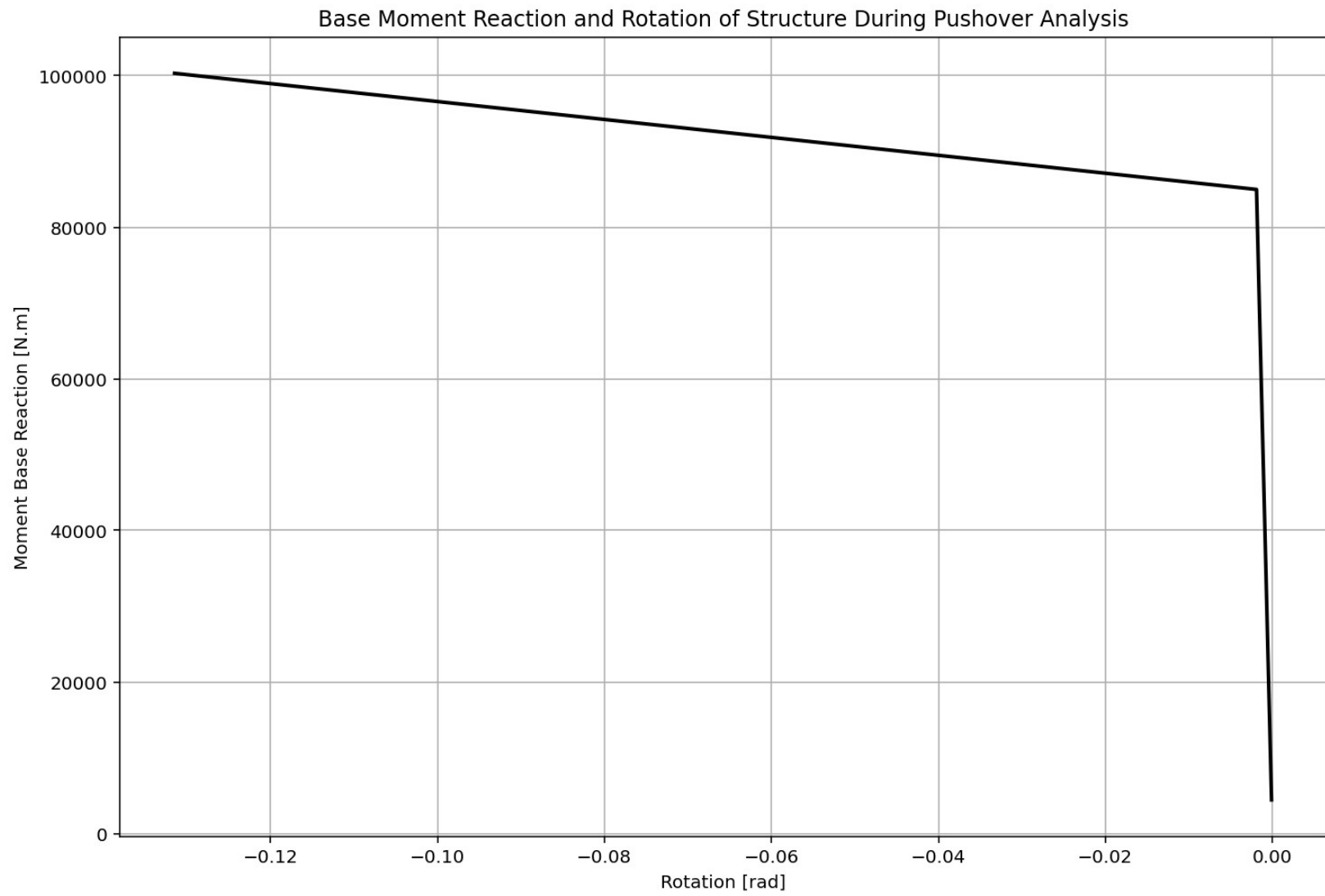
59 %

Base Reaction and Displacement of Structure During Pushover Analysis

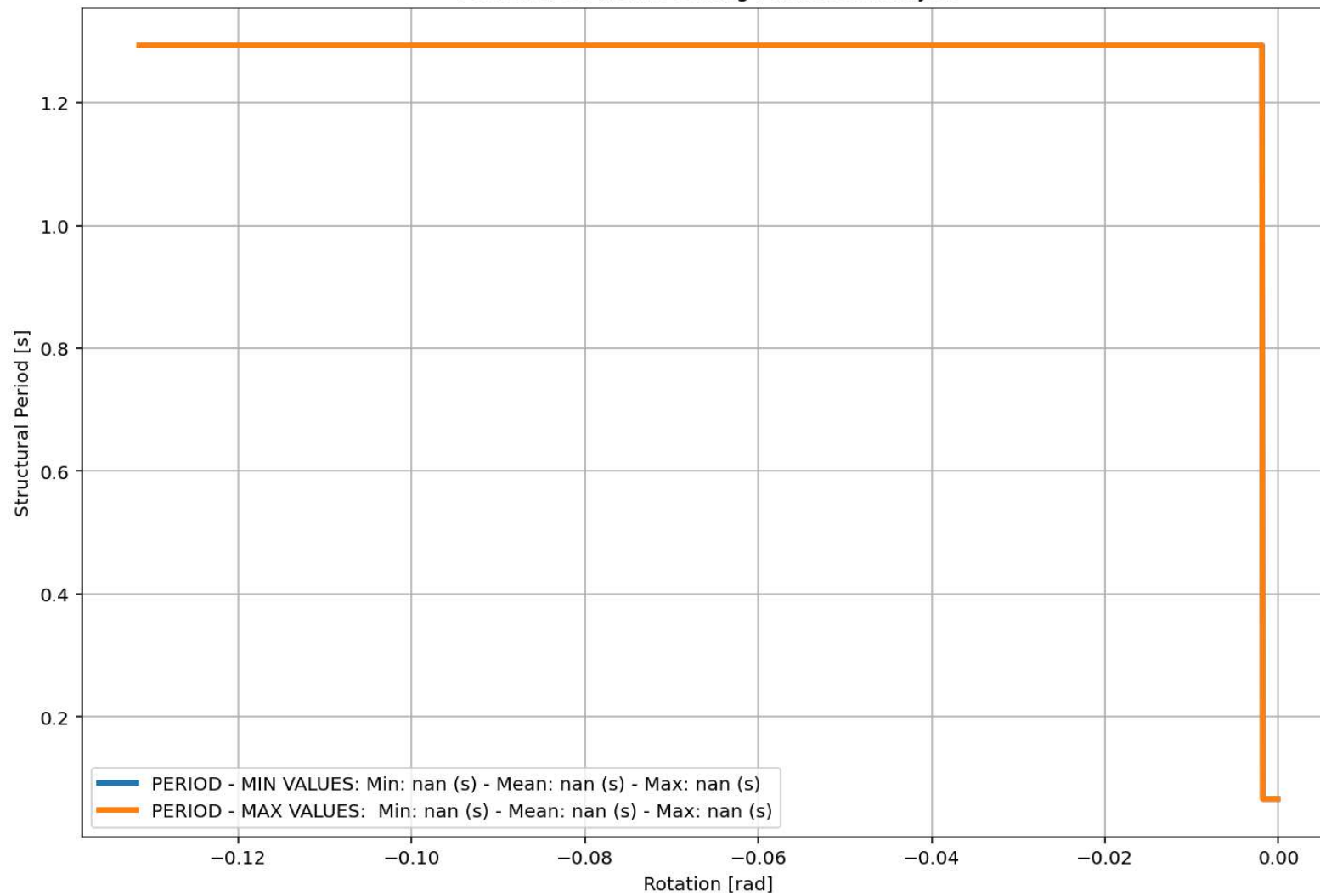


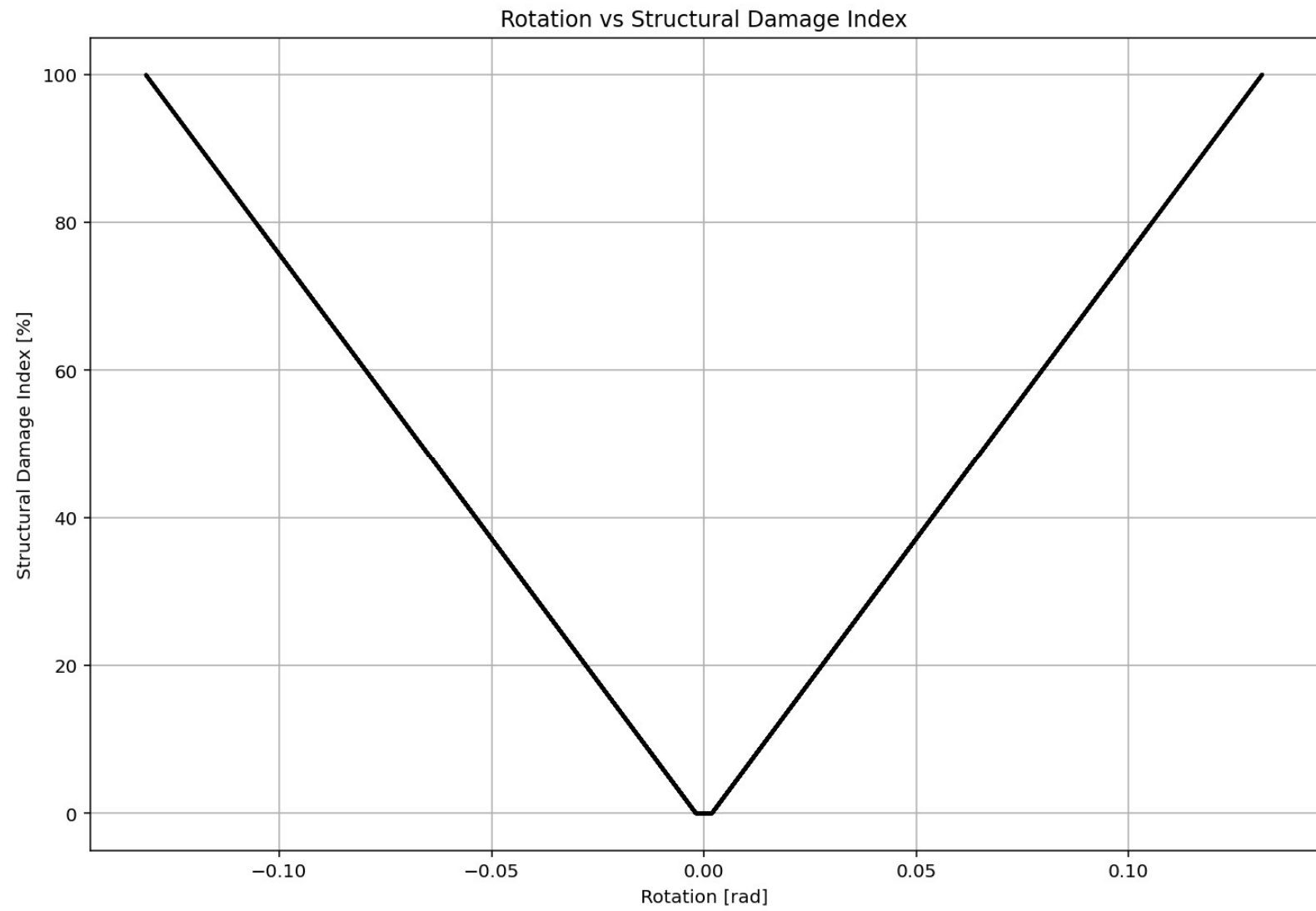
Plots History

Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 3, Col 46 UTF-8 CRLF RW Mem 45%



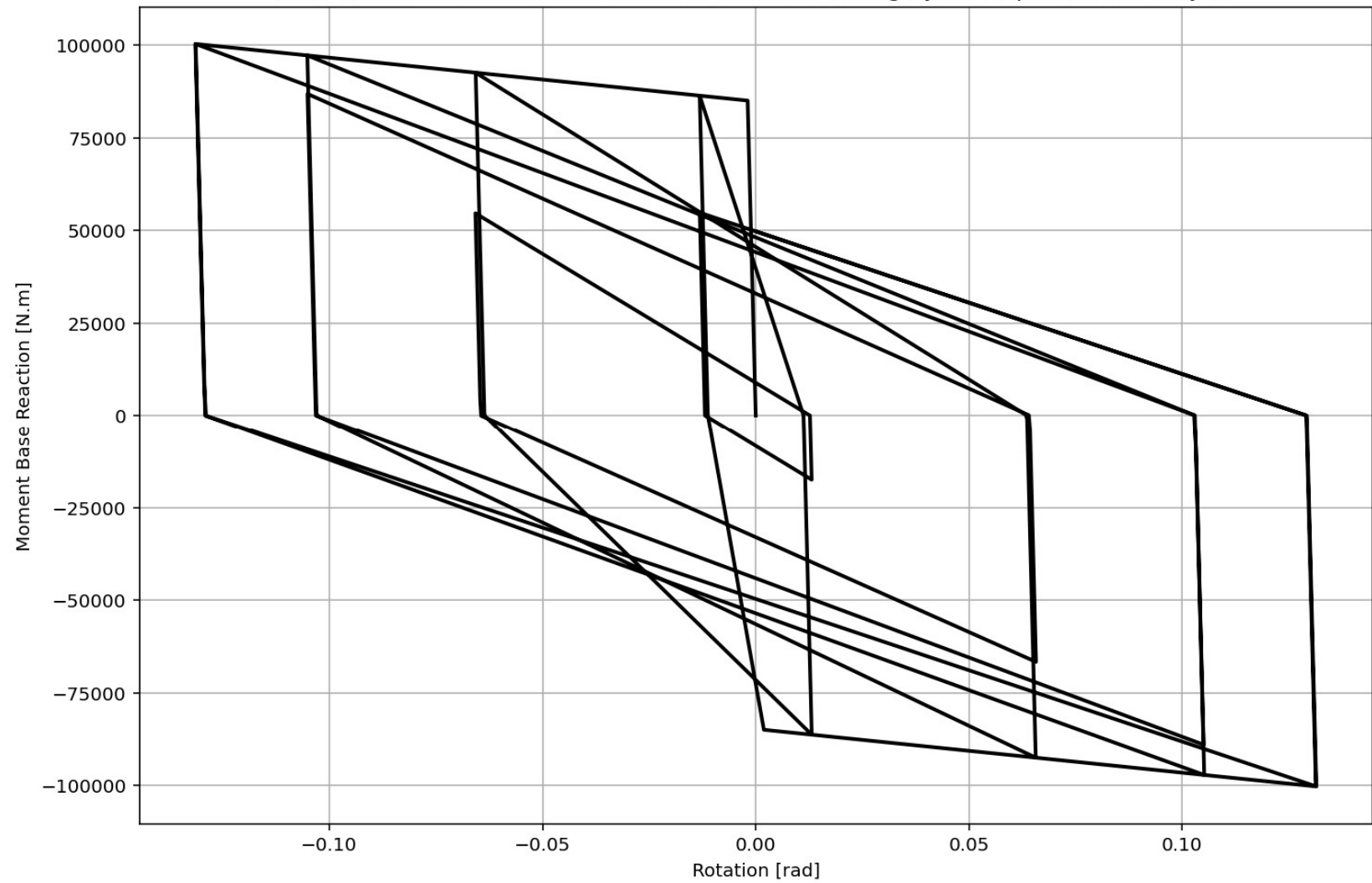
Period of Structure During Pushover Analysis



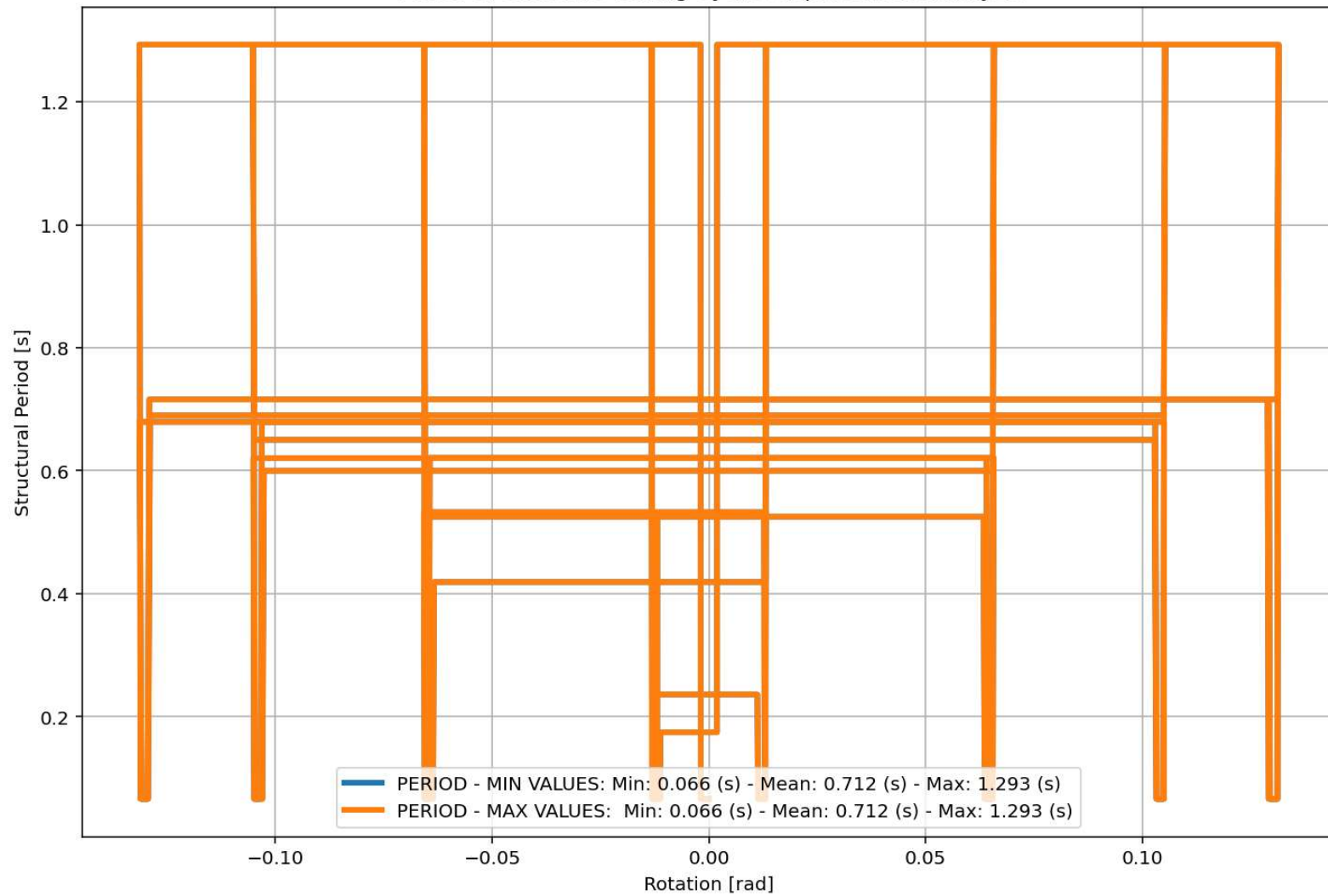


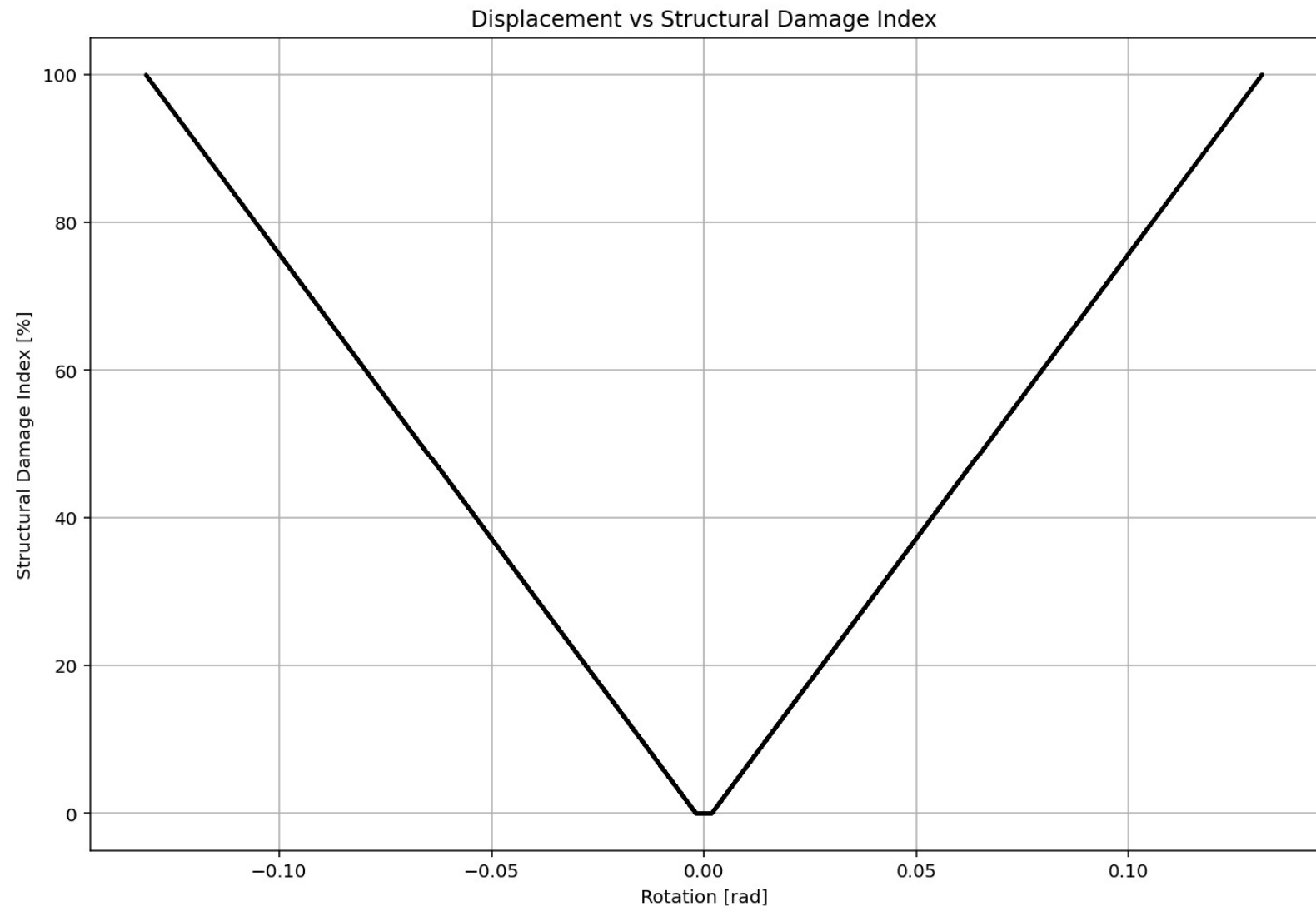
CYCLIC DISPLACEMENT ANALYSIS

Base Moment Reaction and Rotation of Structure During Cyclic-Displacement Analysis

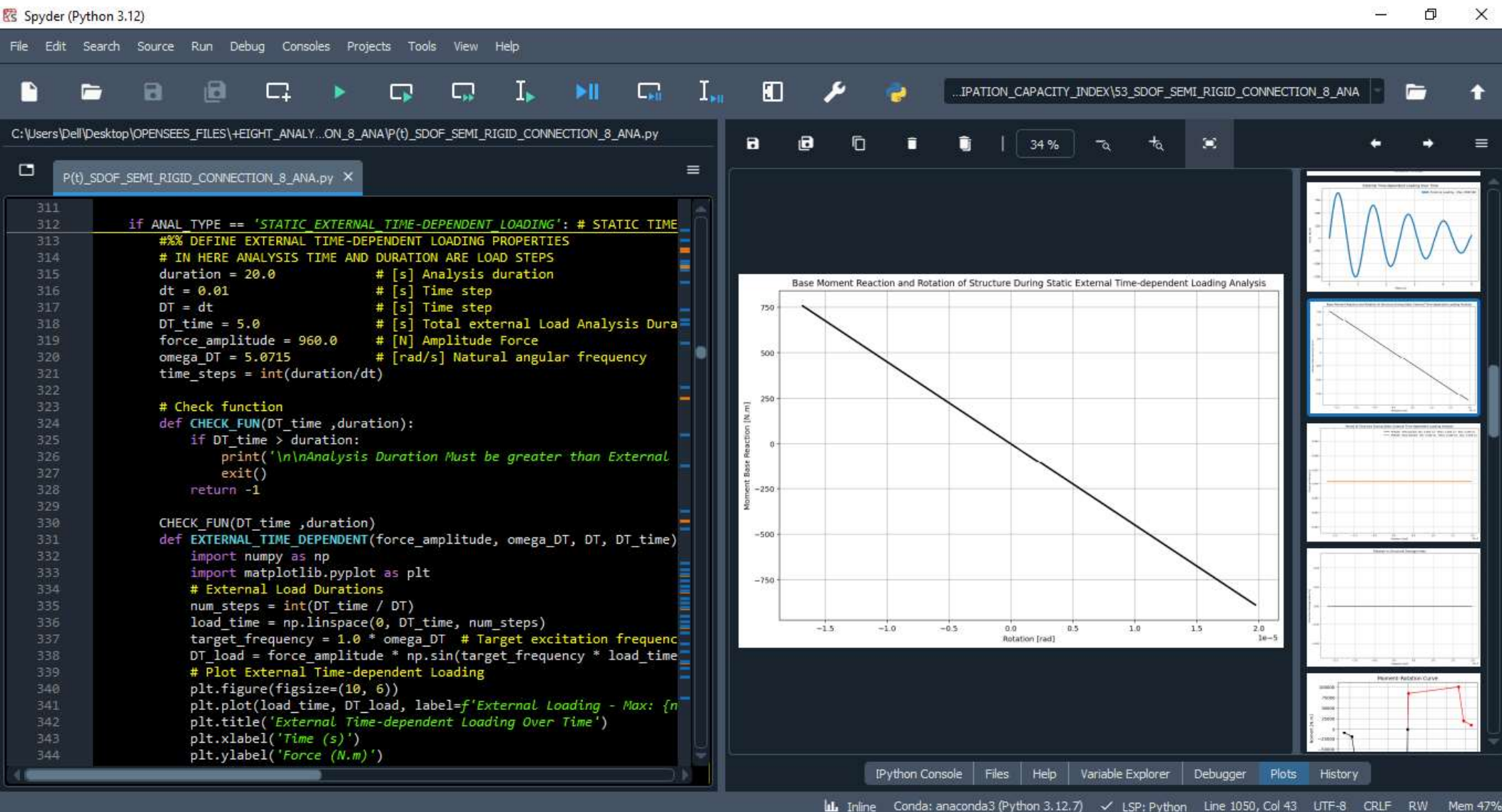


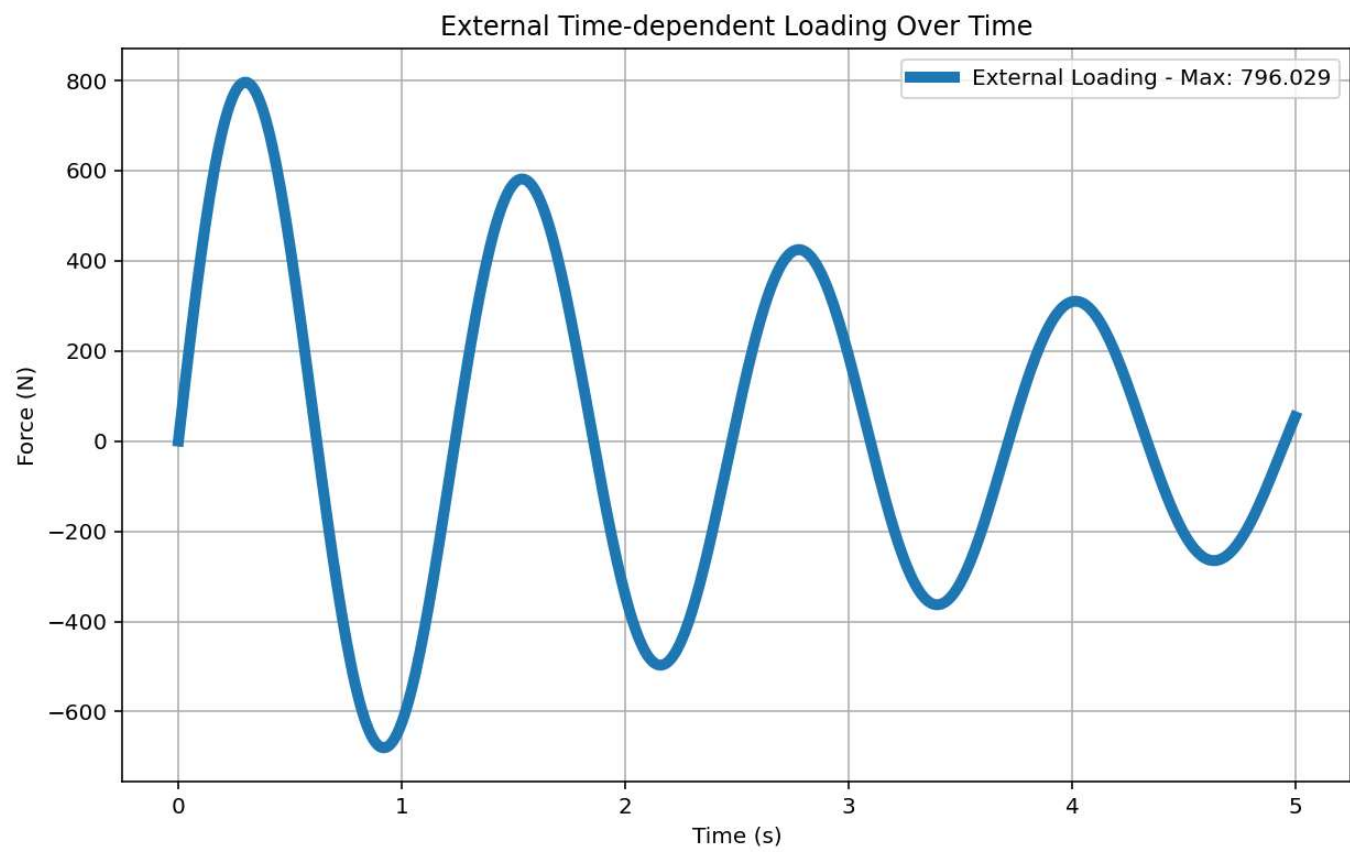
Period of Structure During Cyclic Displacement Analysis



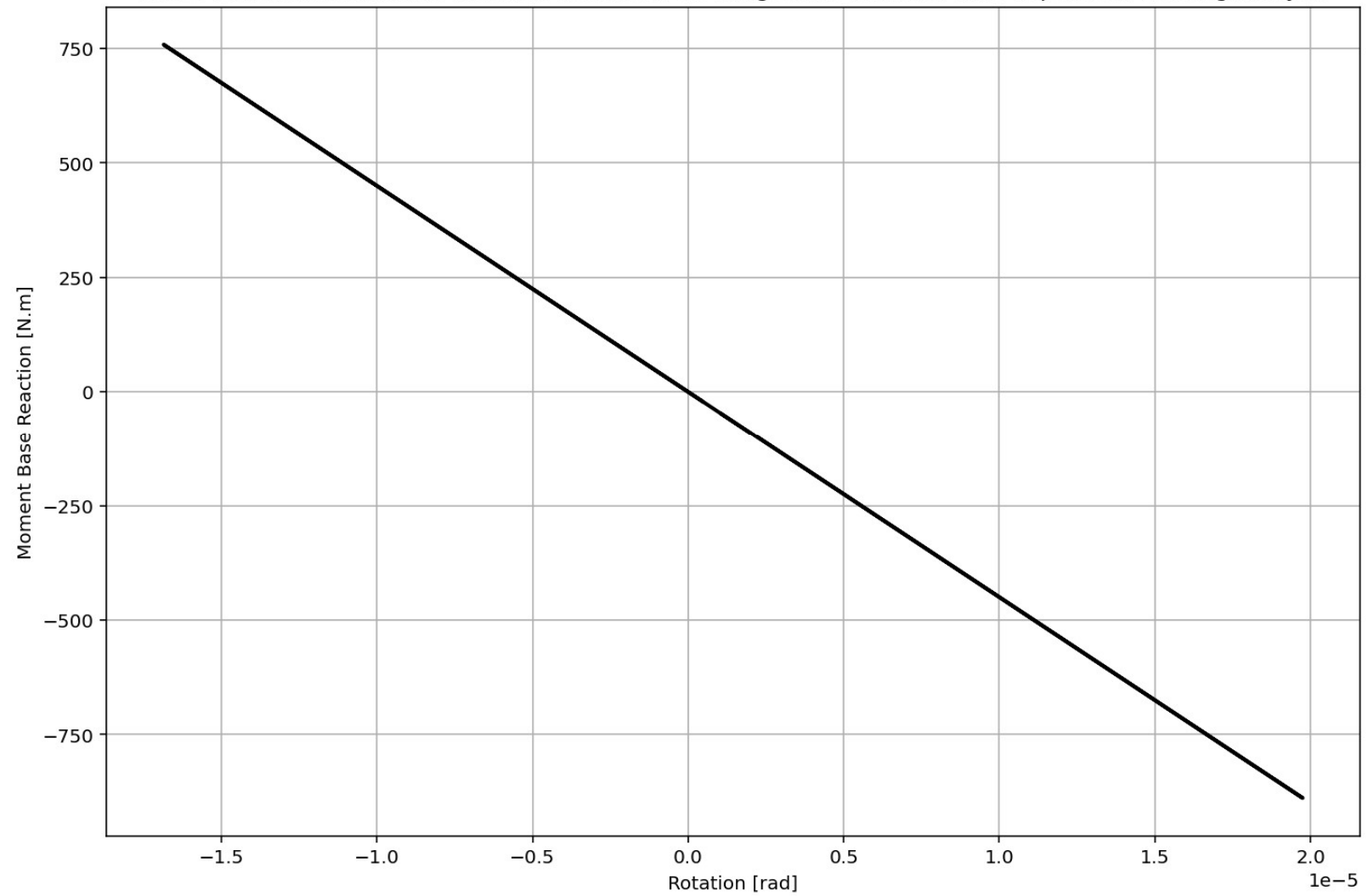


STATIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

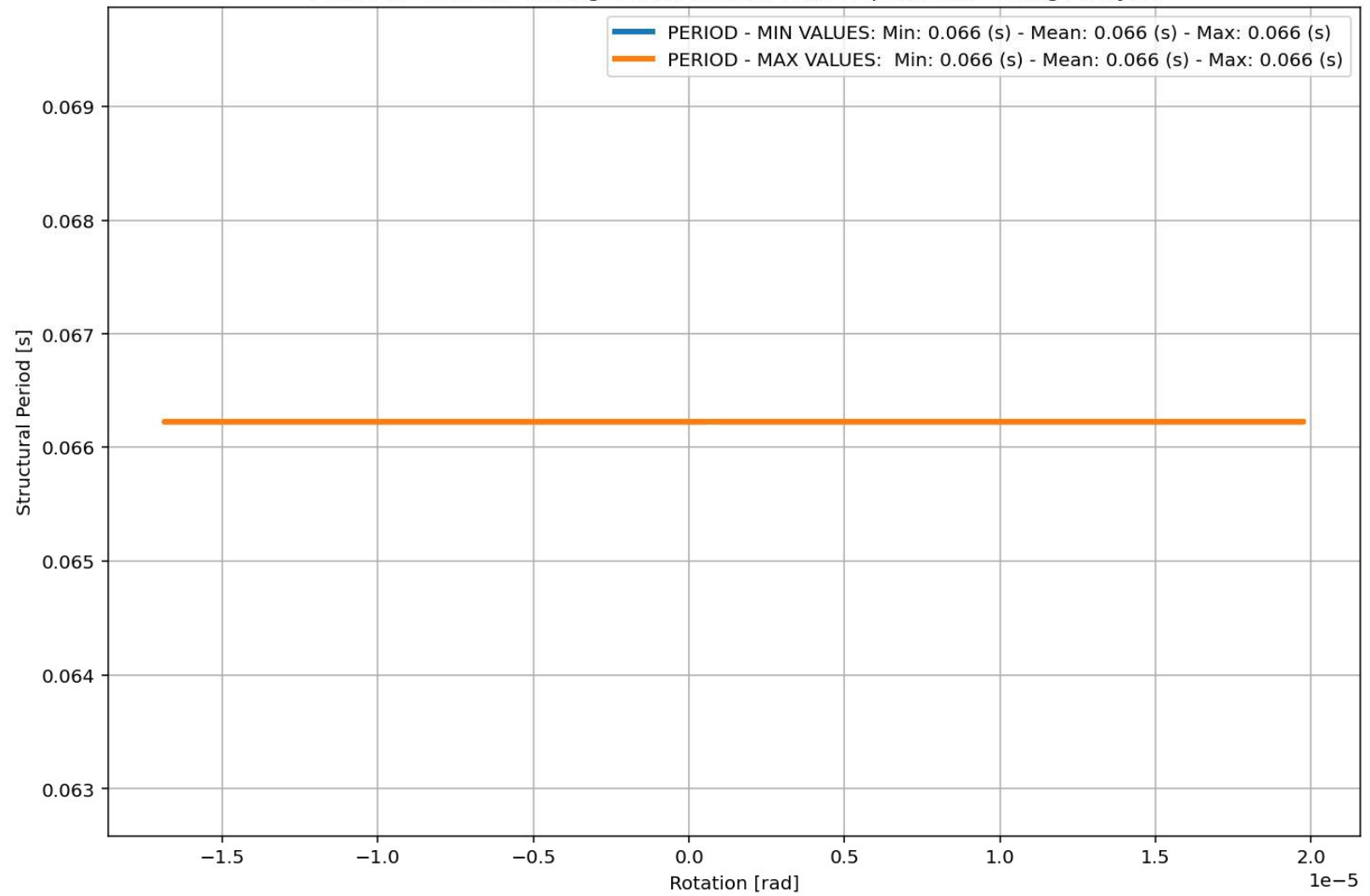


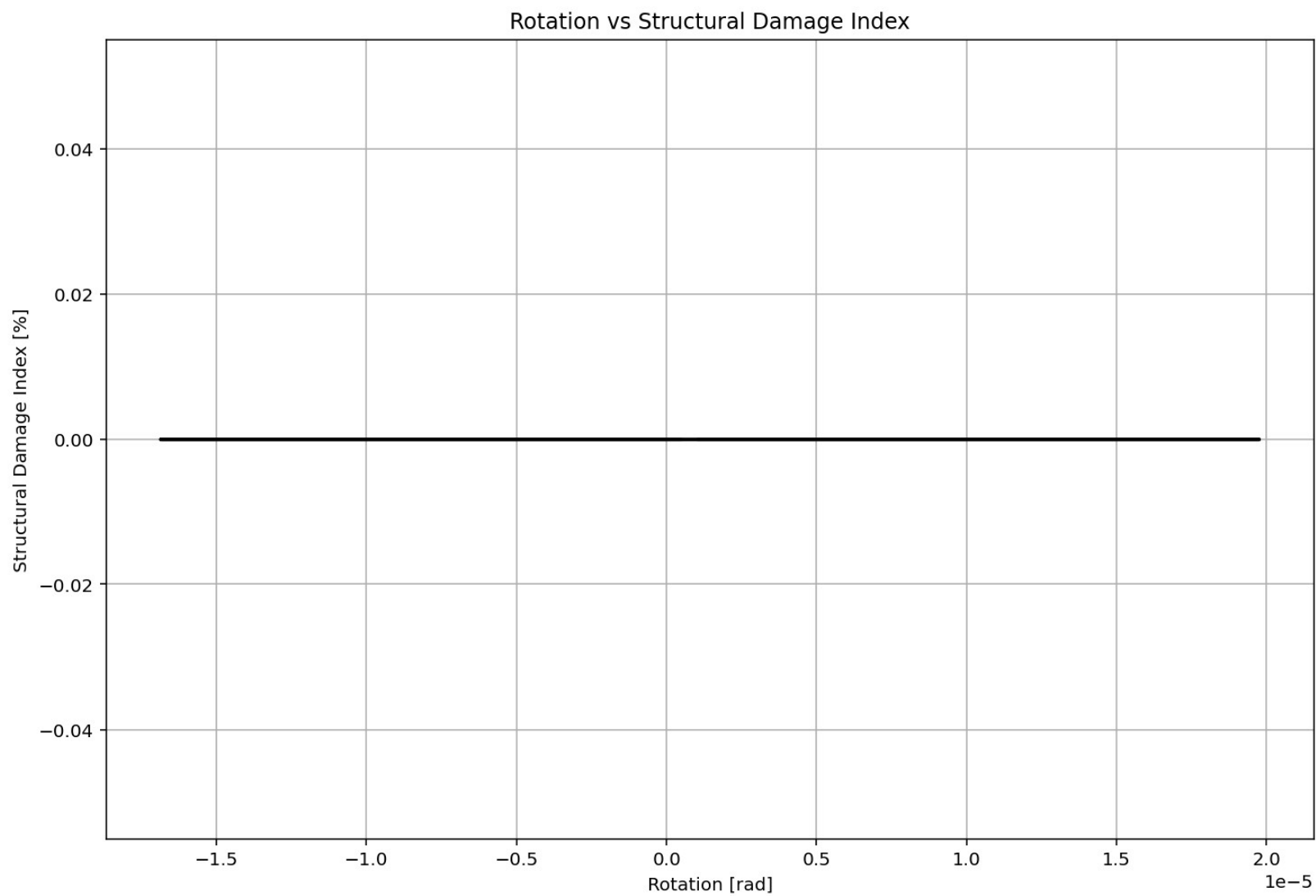


Base Moment Reaction and Rotation of Structure During Static External Time-dependent Loading Analysis



Period of Structure During Static External Time-dependent Loading Analysis



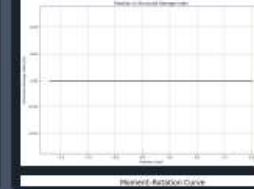
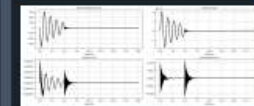
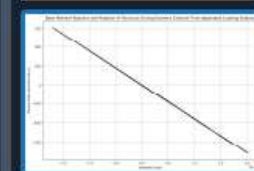
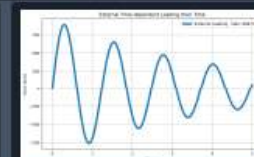
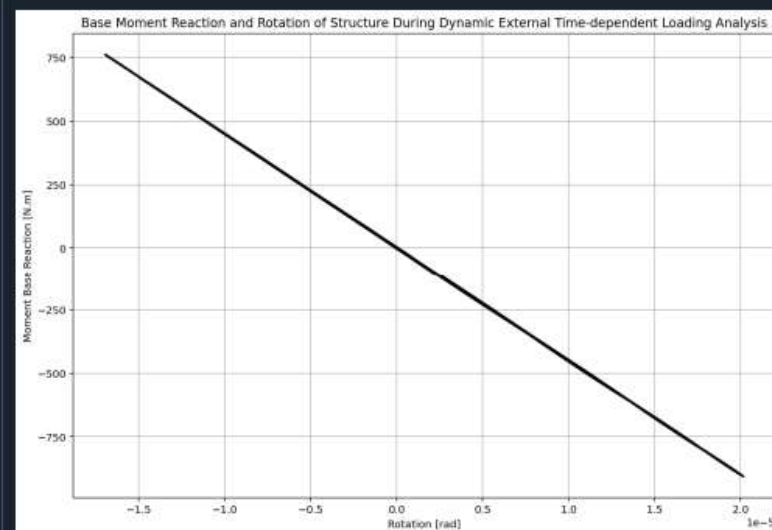


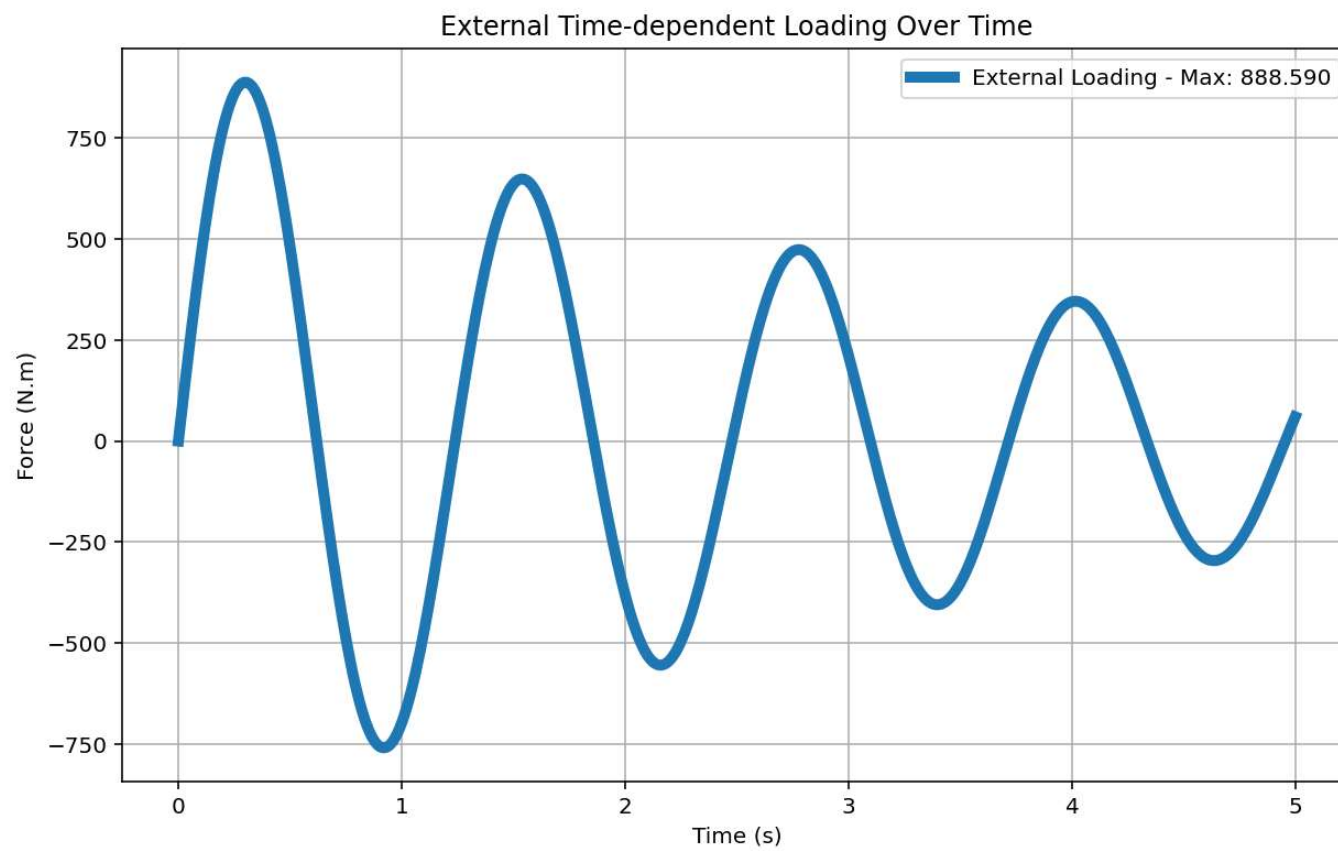
DYNAMIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

```

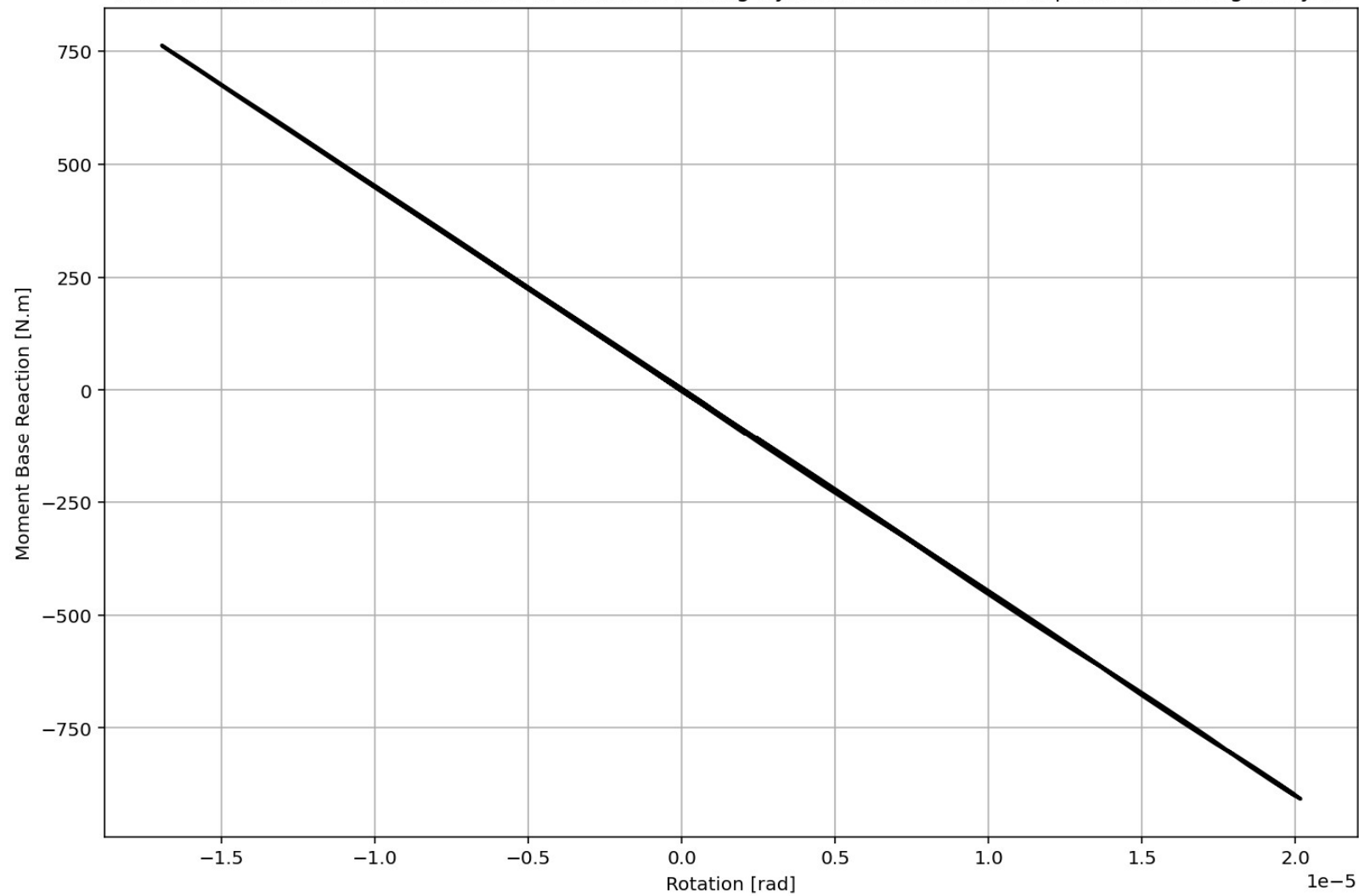
417
418 if ANAL_TYPE == 'DYNAMIC EXTERNAL TIME-DEPENDENT LOADING': # DYNAMIC TI
419     ##% DEFINE EXTERNAL TIME-DEPENDENT LOADING PROPERTIES
420     duration = 20.0 # [s] Analysis duration
421     dt = 0.01 # [s] Time step
422     DT = dt # [s] Time step
423     DT_time = 5.0 # [s] Total external Load Analysis Dura
424     force_amplitude = 960.0 # [N] Amplitude Force
425     omega_DT = 5.0715 # [rad/s] Natural angular frequency
426     DR = 0.03 # Damping Ratio
427
428     # Check function
429     def CHECK_FUN(DT_time ,duration):
430         if DT_time > duration:
431             print('\n\nAnalysis Duration Must be greater than External
432                 exit()
433             return -1
434
435     CHECK_FUN(DT_time ,duration)
436     def EXTERNAL_TIME_DEPENDENT(force_amplitude, omega_DT, DT, DT_time):
437         import numpy as np
438         import matplotlib.pyplot as plt
439         # External Load Durations
440         num_steps = int(DT_time / DT)
441         load_time = np.linspace(0, DT_time, num_steps)
442         target_frequency = 1.0 * omega_DT # Target excitation frequency
443         DT_load = force_amplitude * np.sin(target_frequency * load_time)
444         # Plot External Time-dependent Loading
445         plt.figure(figsize=(10, 6))
446         plt.plot(load_time, DT_load, label=f'External Loading - Max: {n
447         plt.title('External Time-dependent Loading Over Time')
448         plt.xlabel('Time (s)')
449         plt.ylabel('Force (N.m)')
450         plt.grid(True)

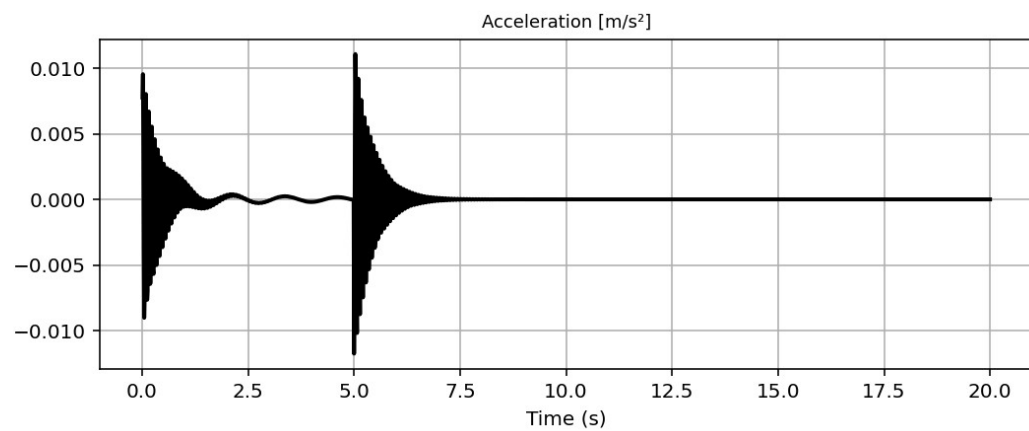
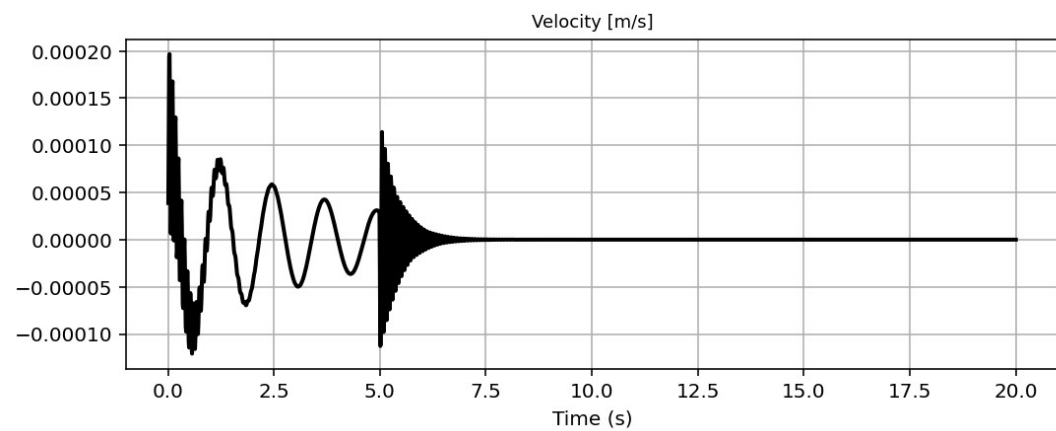
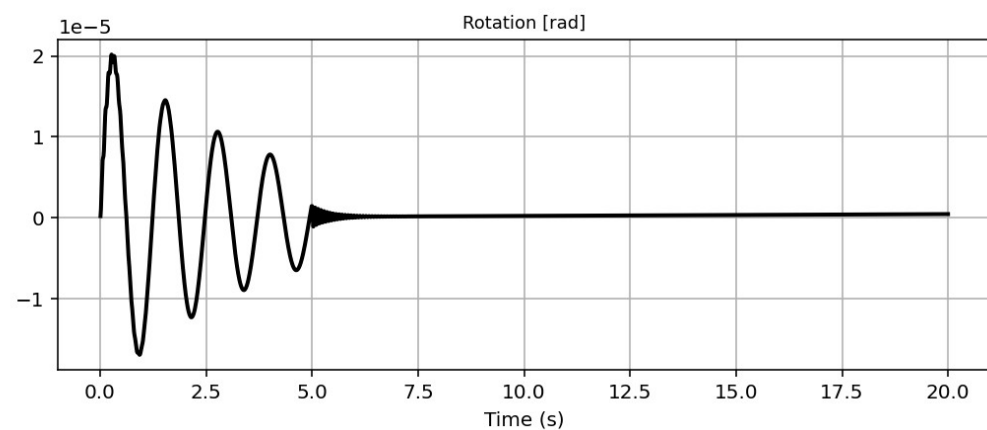
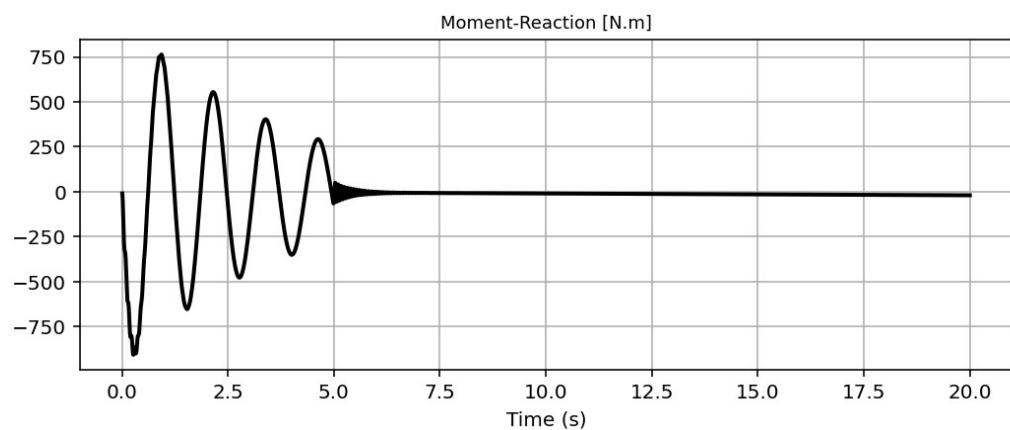
```



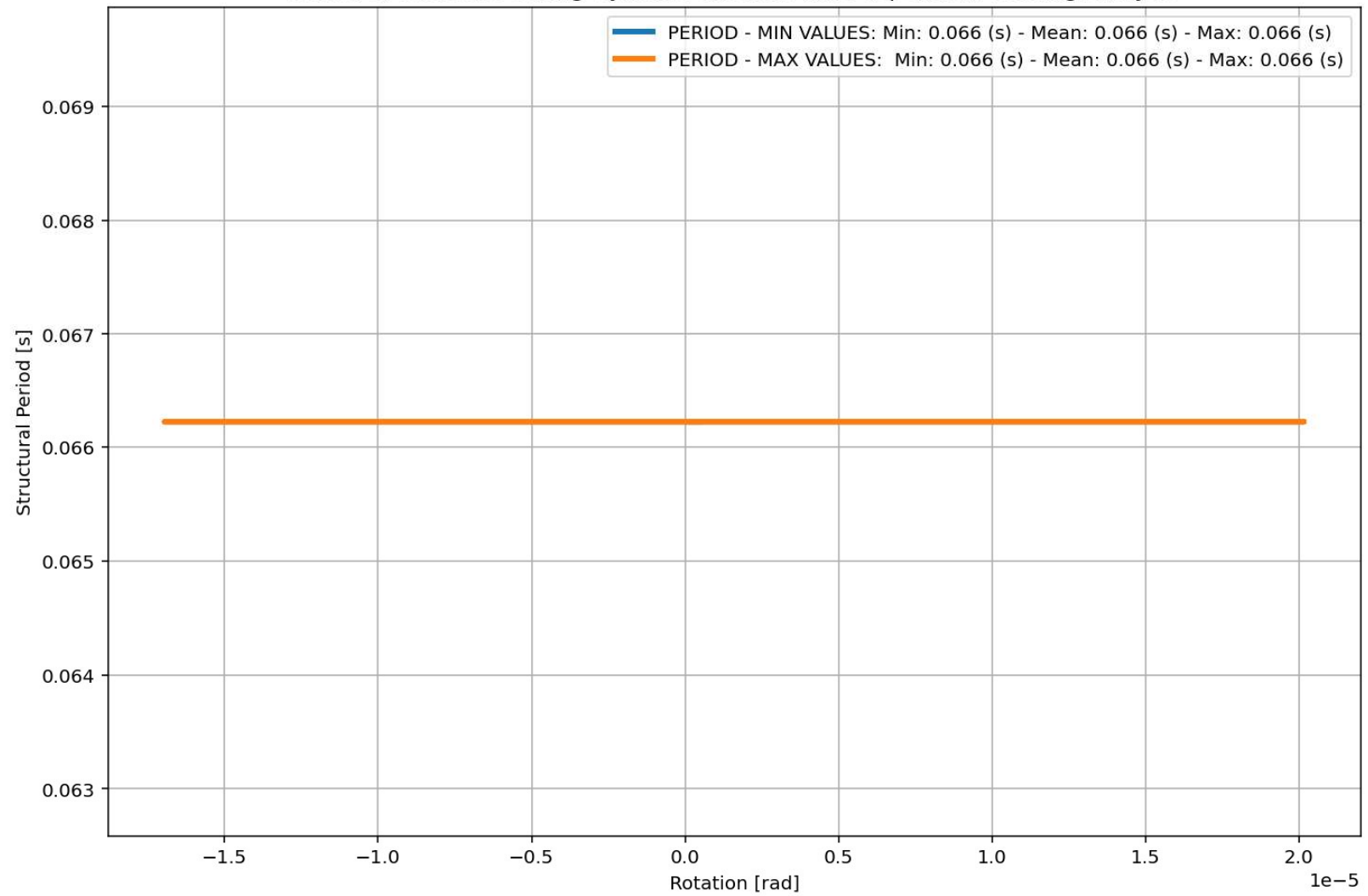


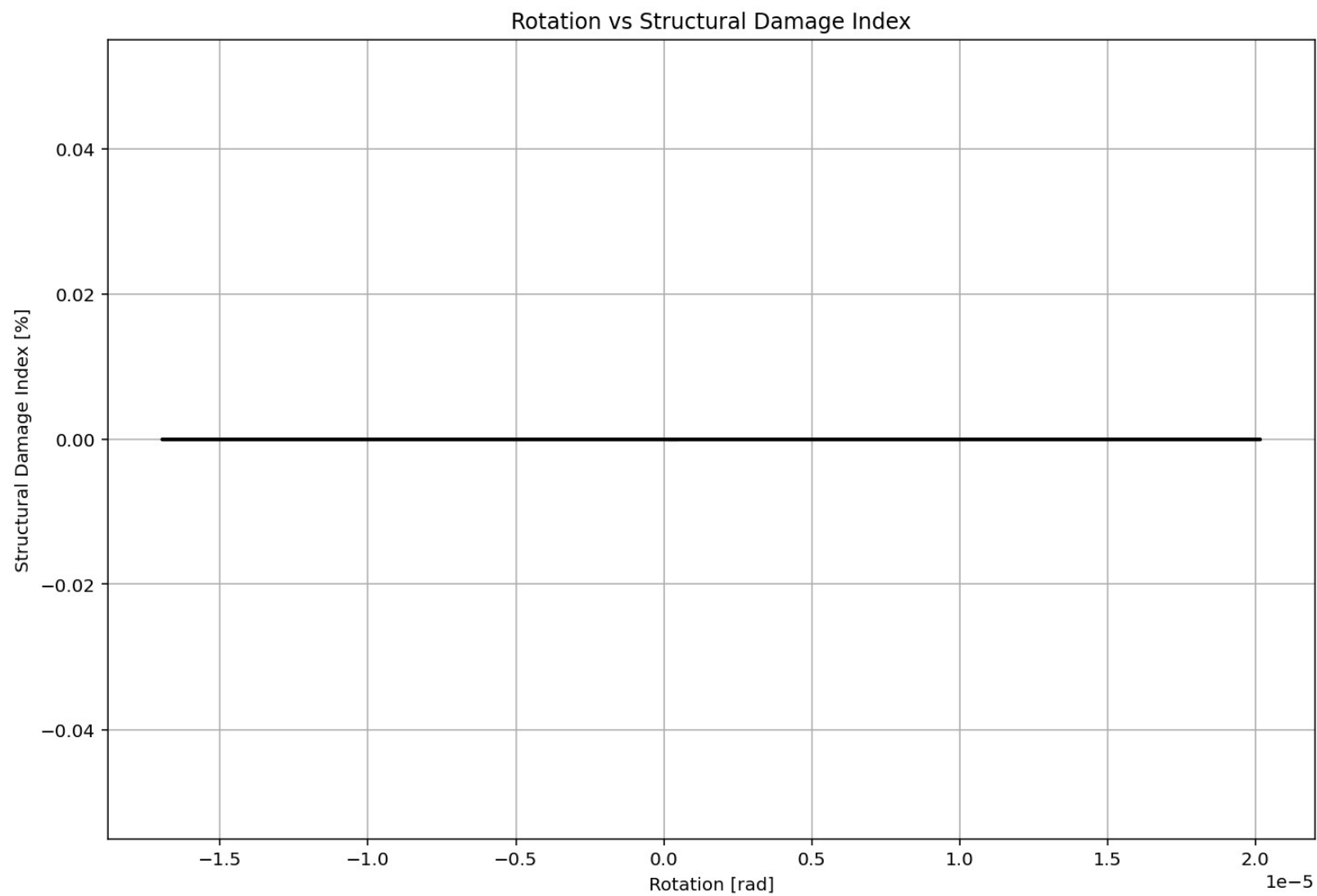
Base Moment Reaction and Rotation of Structure During Dynamic External Time-dependent Loading Analysis





Period of Structure During Dynamic External Time-dependent Loading Analysis

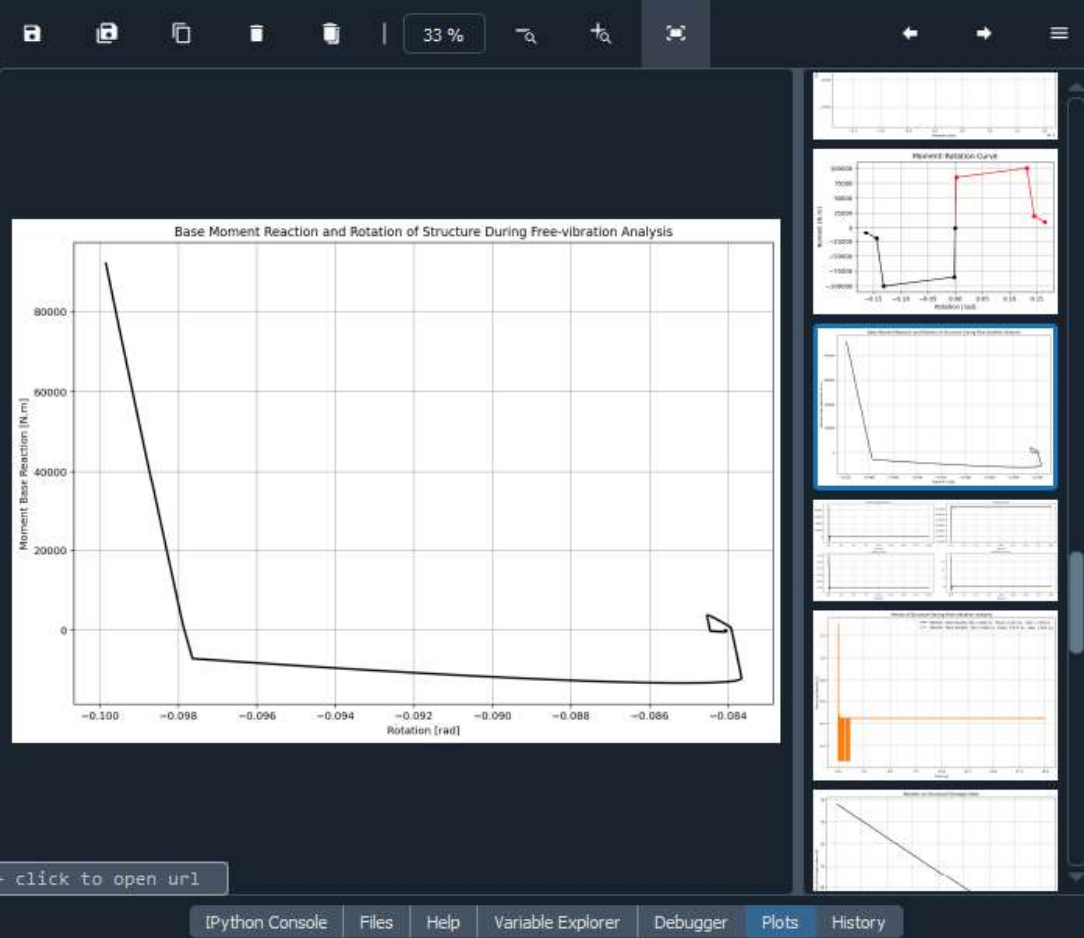




FREE-VIBRATION ANALYSIS

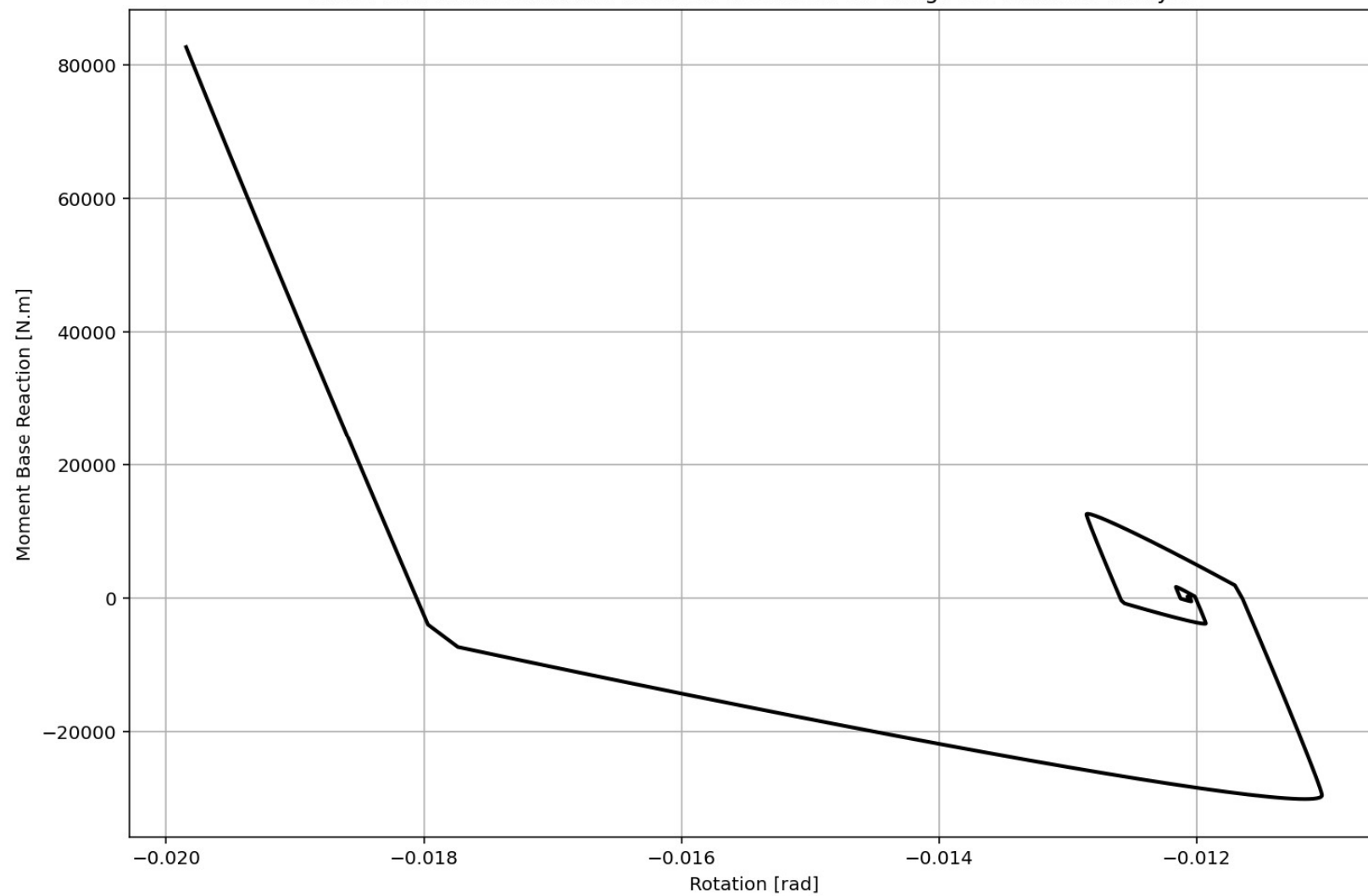
```

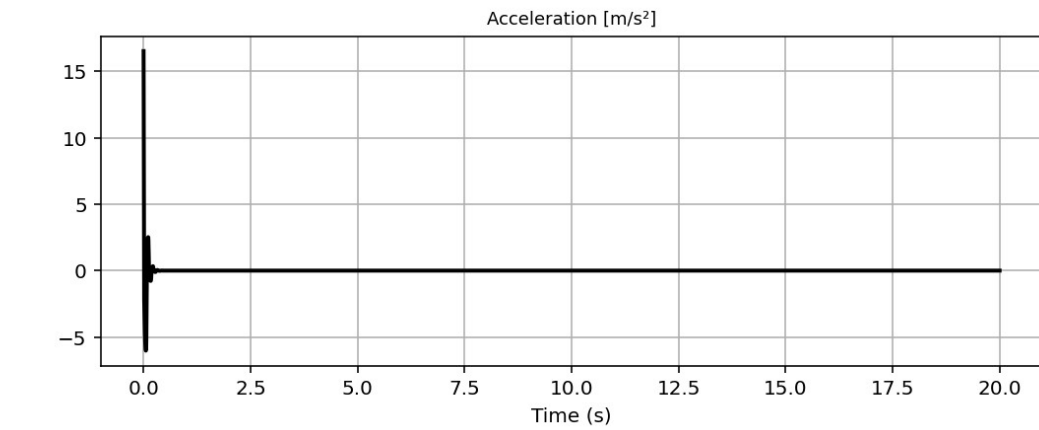
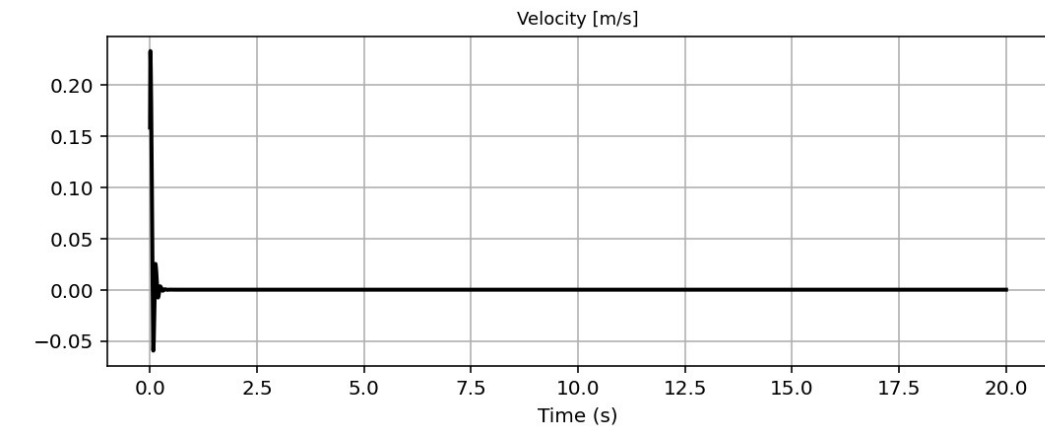
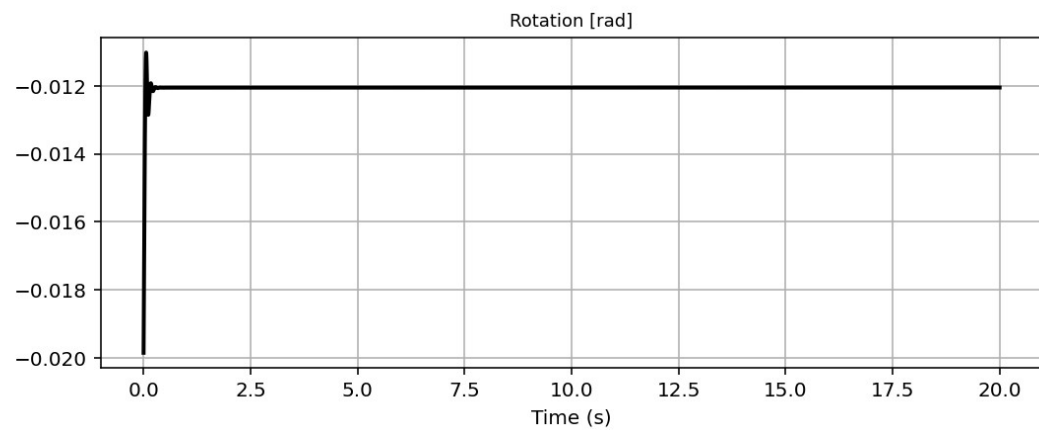
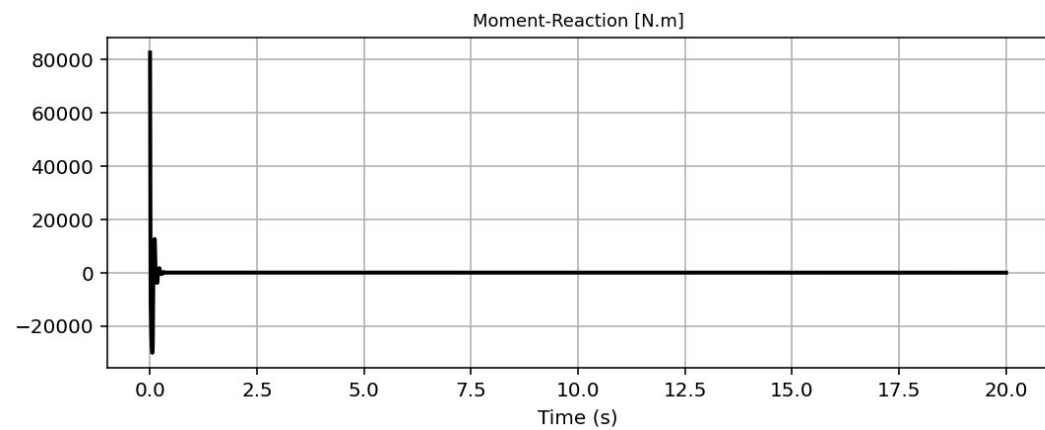
539
540 if ANAL_TYPE == 'FREE-VIBRATION': # DYNAMIC TIME-HISTORY ANALYSIS
541     %% DEFINE PARAMETERS FOR FREE-VIBRATION ANALYSIS
542     u0 = -0.10 # [m] Initial displacement
543     v0 = 0.15 # [m/s] Initial velocity
544     a0 = 0.065 # [m/s^2] Initial acceleration
545     IU = True # Free Vibration with Initial Di
546     IV = True # Free Vibration with Initial Ve
547     IA = True # Free Vibration with Initial Ac
548     duration = 20.0 # [s] Analysis duration
549     dt = 0.001 # [s] Time step
550     DR = 0.03 # Damping Ratio
551
552     if IU == True:
553         # Define initial displacment
554         ops.setNodeDisp(center_node, 1, u0, '-commit') # INFO LINK: htt
555     if IV == True:
556         # Define initial velocity
557         ops.setNodeVel(center_node, 1, v0, '-commit') # INFO LINK: htt
558     if IA == True:
559         # Define initial acceleration
560         ops.setNodeAccel(center_node, 1, a0, '-commit') # INFO LINK: ht
561
562     ops.constraints('Plain')
563     ops.numberer('Plain')
564     ops.system('BandGeneral')
565     ops.test('NormDispIncr', MAX_TOLERANCE, MAX_ITERATIONS) # INFO LINK
566     #ops.integrator('CentralDifference') # JUST FOR LINEAR ANALYSIS -
567     alpha=0.5; beta=0.25;
568     ops.integrator('Newmark', alpha, beta) # INFO LINK: https://opensee
569     #alpha=2/3; gamma=1.5-alpha; gamma=1.5-alpha; beta=(2-alpha)**2/4;
570     #ops.integrator('HHT', alpha, gamma, beta) # INFO LINK: https://ope
571     ops.algorithm('Newton') # INFO LINK: https://openseespydoc.readth
572     ops.analysis('Transient') # INFO LINK: https://openseespydoc.readth
    
```



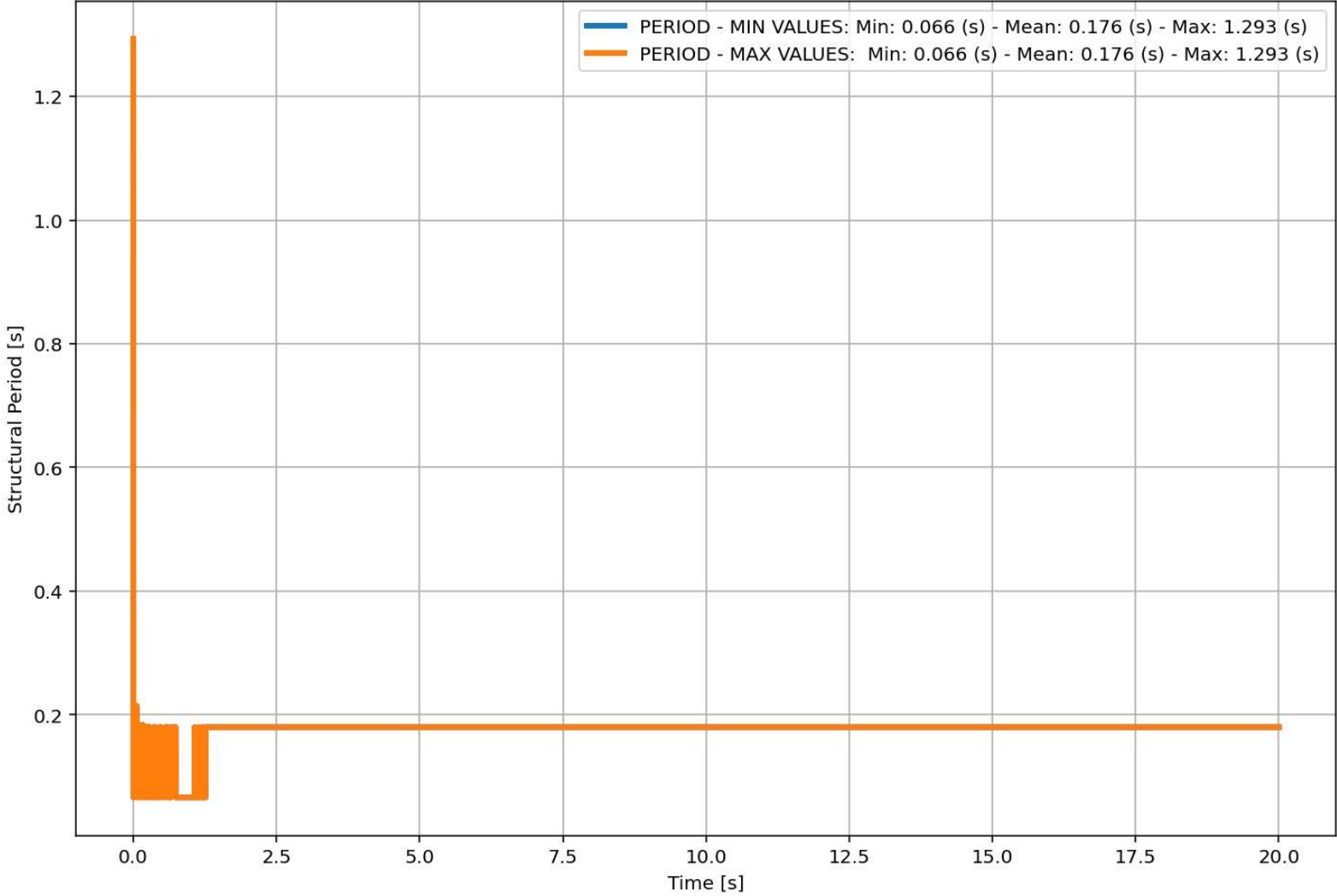
Ctrl + click to open url

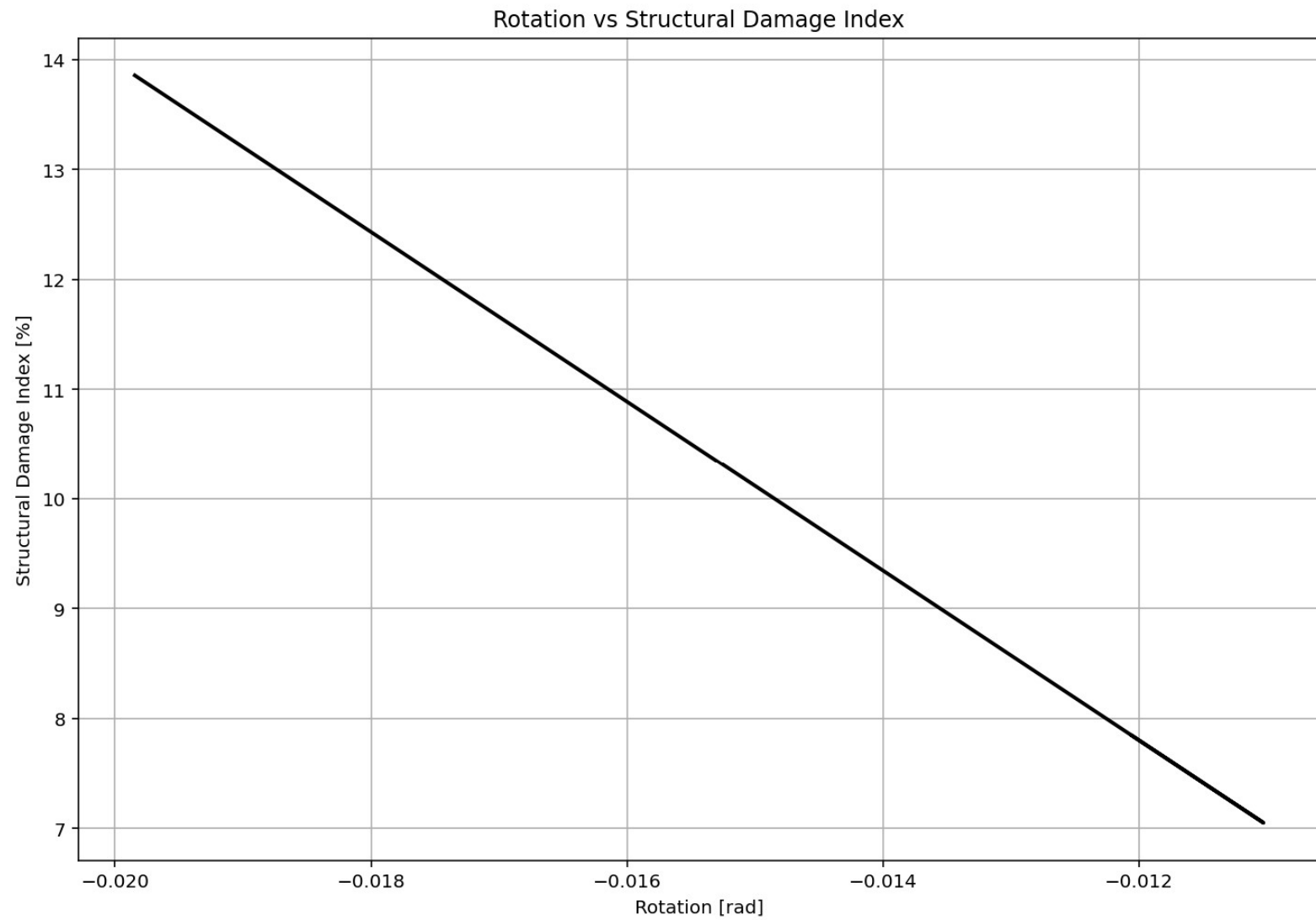
Base Moment Reaction and Rotation of Structure During Free-vibration Analysis



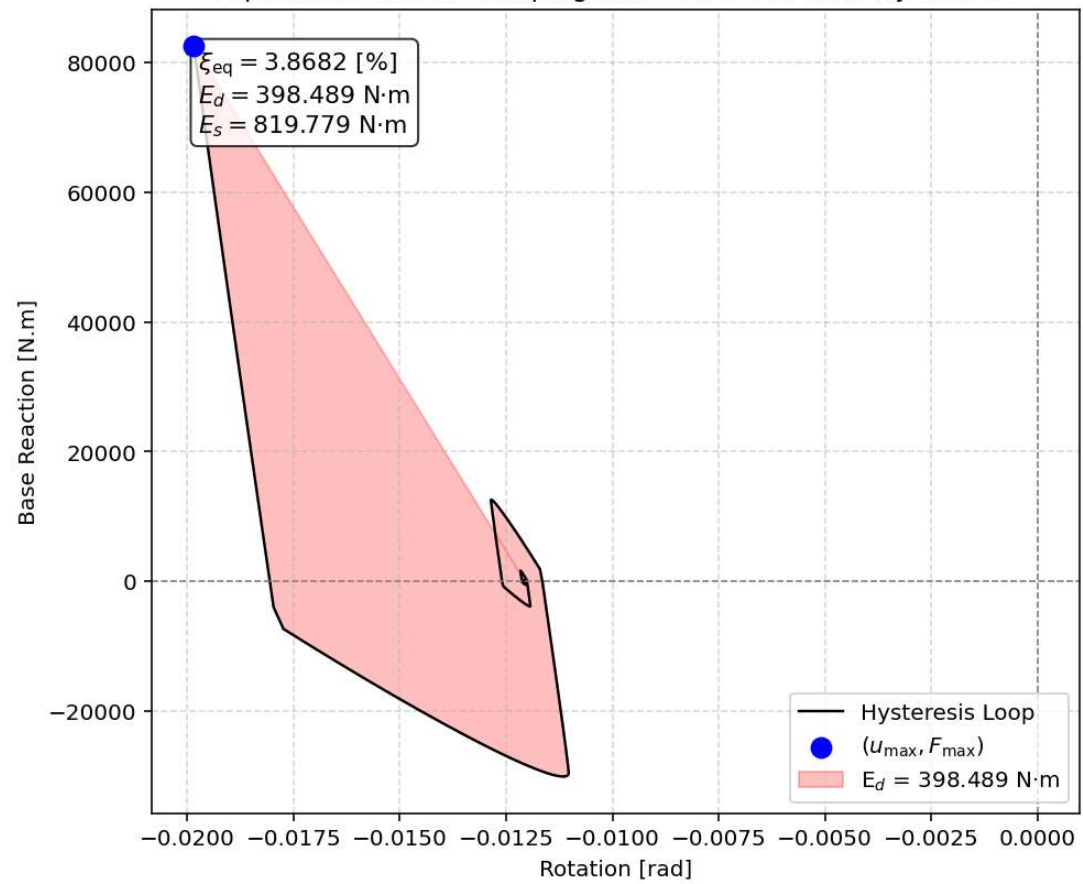


Period of Structure During Free-vibration Analysis





Equivalent viscous damping ratio - Free-vibration Hysteresis



SEISMIC ANALYSIS

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...ON_8_ANA\P(t)_SDOF_SEMI_RIGID_CONNECTION_8_ANA.py

P(t)_SDOF_SEMI_RIGID_CONNECTION_8_ANA.py

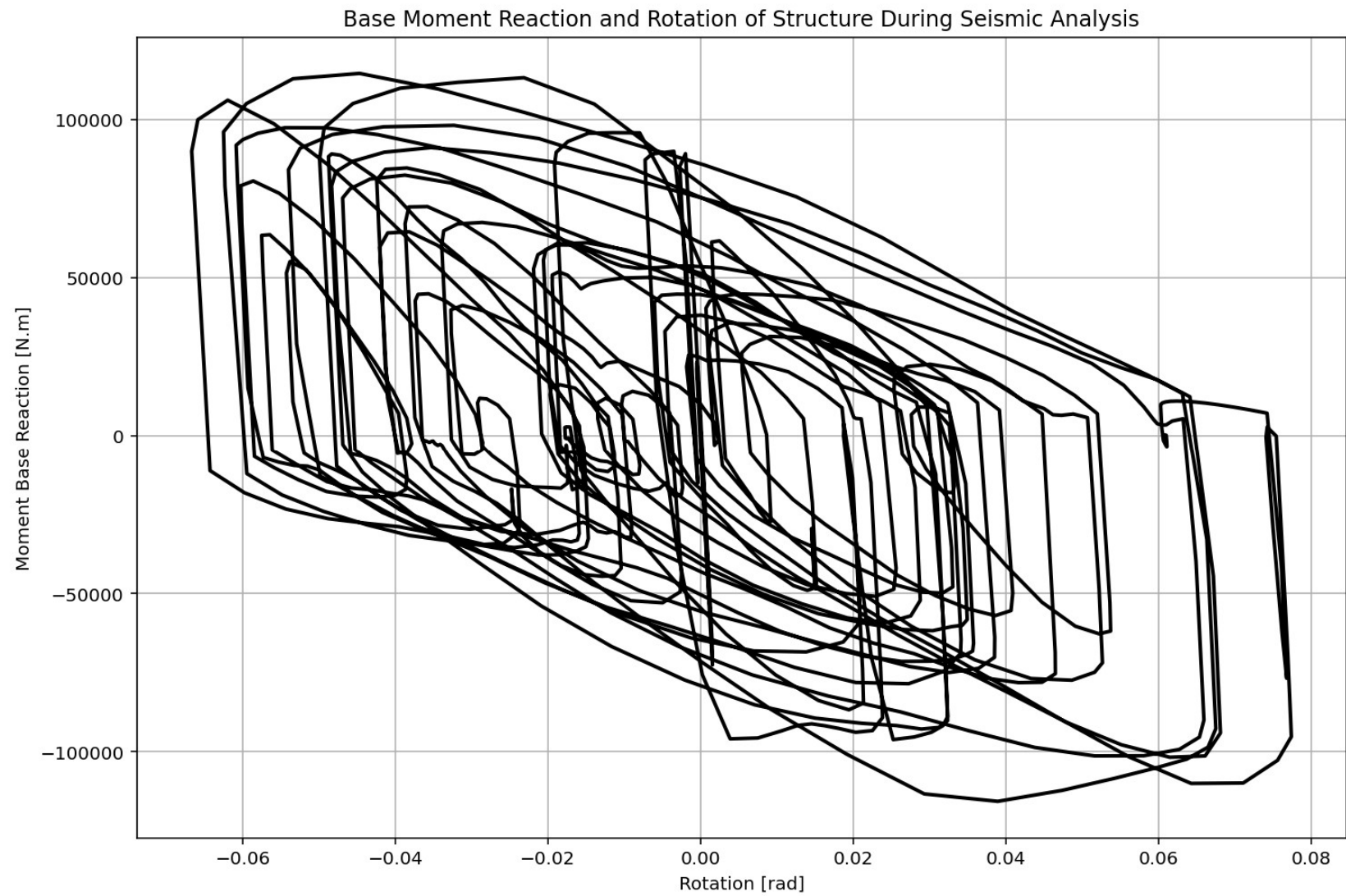
```
614 |
615 | if ANAL_TYPE == 'SEISMIC': # DYNAMIC TIME-HISTORY ANALYSIS
616 |     %% DEFINE PARAMETERS FOR SEISMIC ANALYSIS
617 |     duration = 20.0           # [s] Analysis duration
618 |     dt = 0.01                # [s] Time step
619 |     GMfact = 9810            # [mm/s²] standard acceleration o
620 |     SSF_X = 0.10             # Seismic Acceleration Scale Fac
621 |     SSF_Y = 0.10             # Seismic Acceleration Scale Fac
622 |     iv0_X = 0.0005           # [mm/s] Initial velocity applie
623 |     iv0_Y = 0.0005           # [mm/s] Initial velocity applie
624 |     st_iv0 = 0.0             # [s] Initial velocity applied s
625 |     SEI = 'X'                # Seismic Direction
626 |     DR = 0.03                # Damping ratio
627 |
628 |     # Define time series for input motion (Acceleration time history)
629 |     if SEI == 'X':
630 |         SEISMIC_TAG_01 = 100
631 |         ops.timeSeries('Path', SEISMIC_TAG_01, '-dt', dt, '-filePath',
632 |             # Define load patterns
633 |             # pattern UniformExcitation $patternTag $dof -accel $tsTag <-ve
634 |         ops.pattern('UniformExcitation', SEISMIC_TAG_01, 1, '-accel', S
635 |     if SEI == 'Y':
636 |         SEISMIC_TAG_02 = 200
637 |         ops.timeSeries('Path', SEISMIC_TAG_02, '-dt', dt, '-filePath',
638 |         ops.pattern('UniformExcitation', SEISMIC_TAG_02, 2, '-accel', S
639 |     if SEI == 'XY':
640 |         SEISMIC_TAG_01 = 100
641 |         ops.timeSeries('Path', SEISMIC_TAG_01, '-dt', dt, '-filePath',
642 |         # Define load patterns
643 |         # pattern UniformExcitation $patternTag $dof -accel $tsTag <-ve
644 |         ops.pattern('UniformExcitation', SEISMIC_TAG_01, 1, '-accel', S
645 |         SEISMIC_TAG_02 = 200
646 |         ops.timeSeries('Path', SEISMIC_TAG_02, '-dt', dt, '-filePath',
647 |         ops.pattern('UniformExcitation', SEISMIC_TAG_02, 2, '-accel', S
```

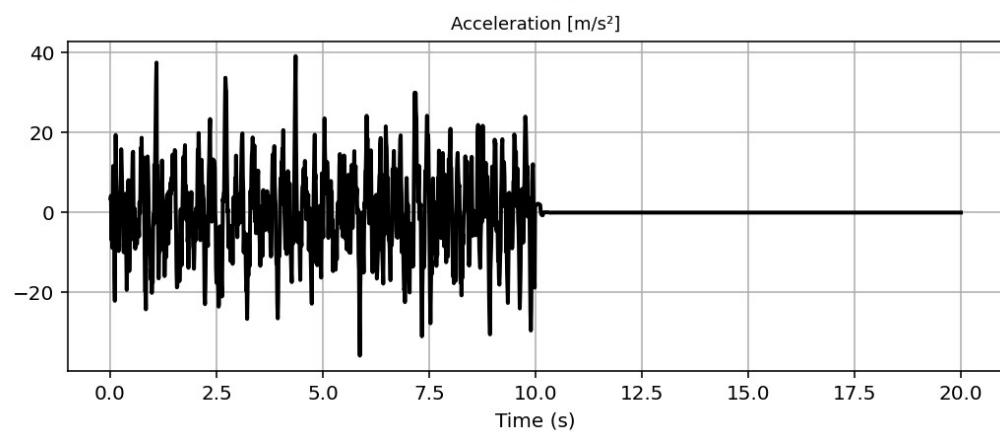
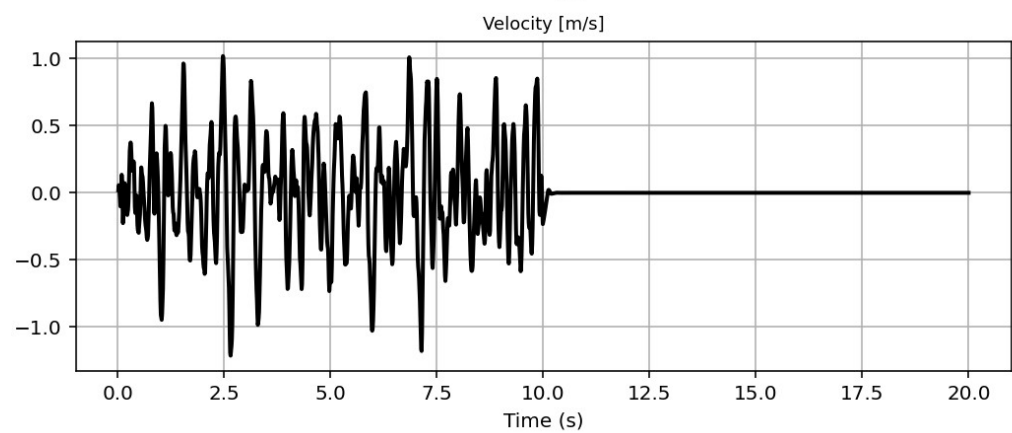
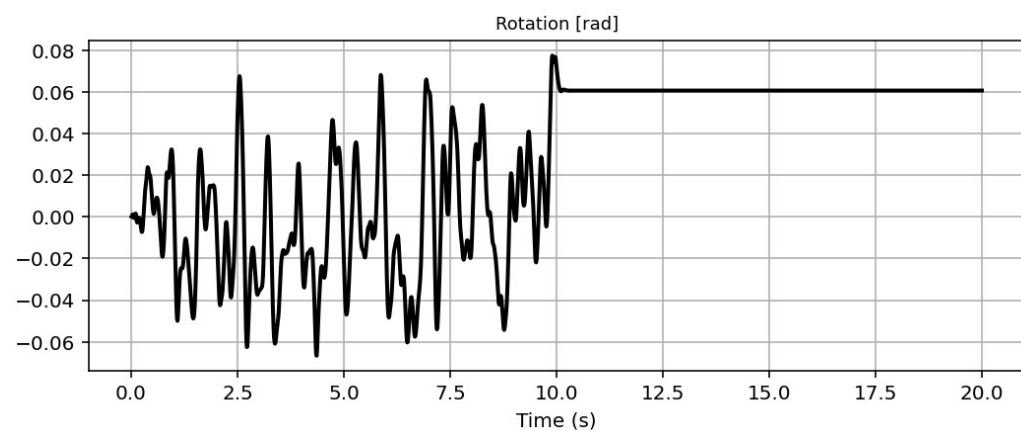
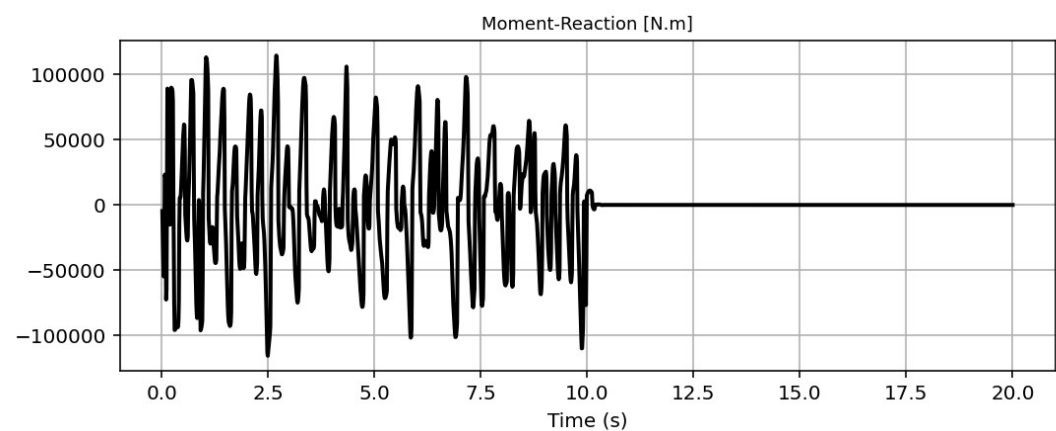
Base Moment Reaction and Rotation of Structure During Seismic Analysis

33 %

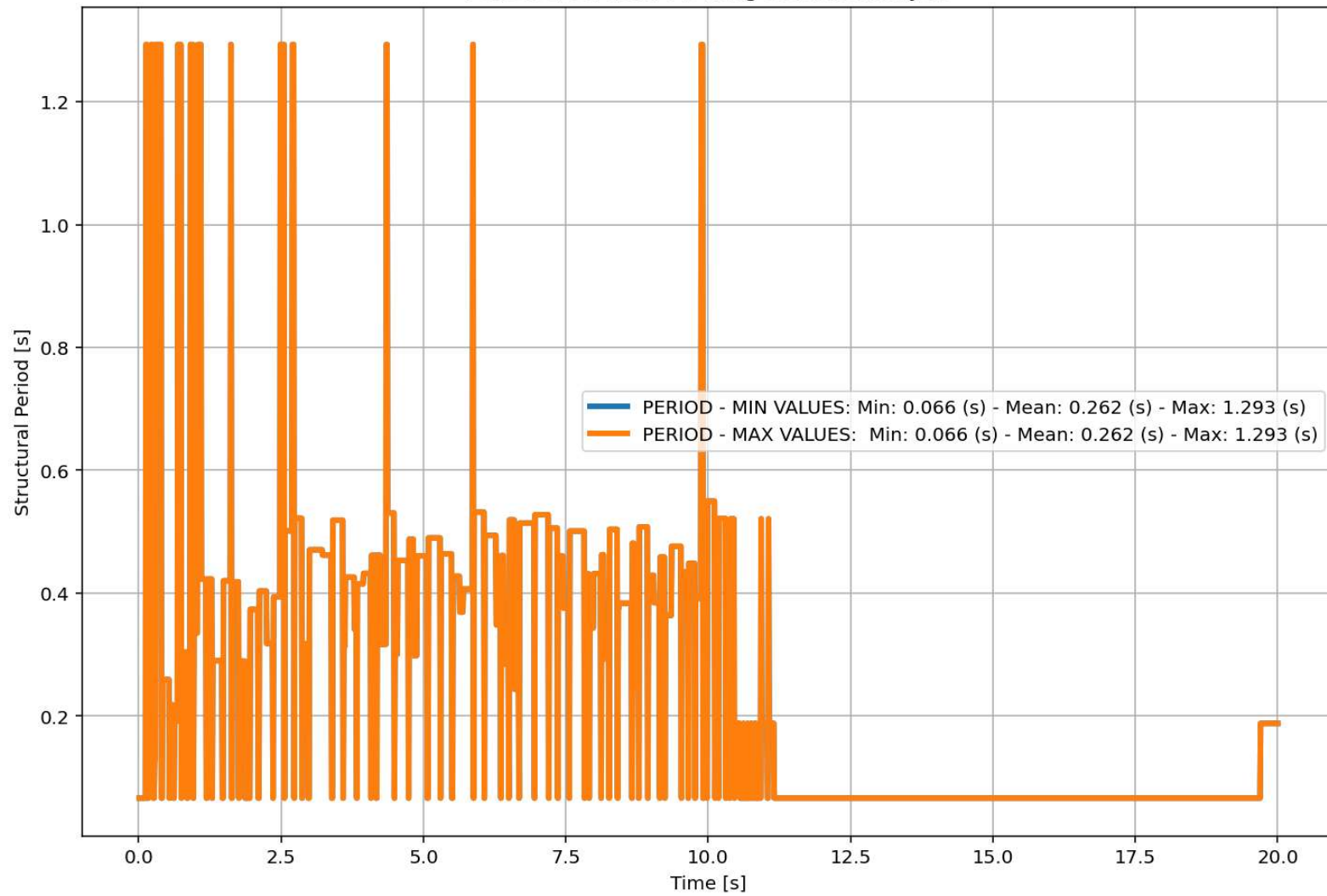
IPython Console Files Help Variable Explorer Debugger Plots History

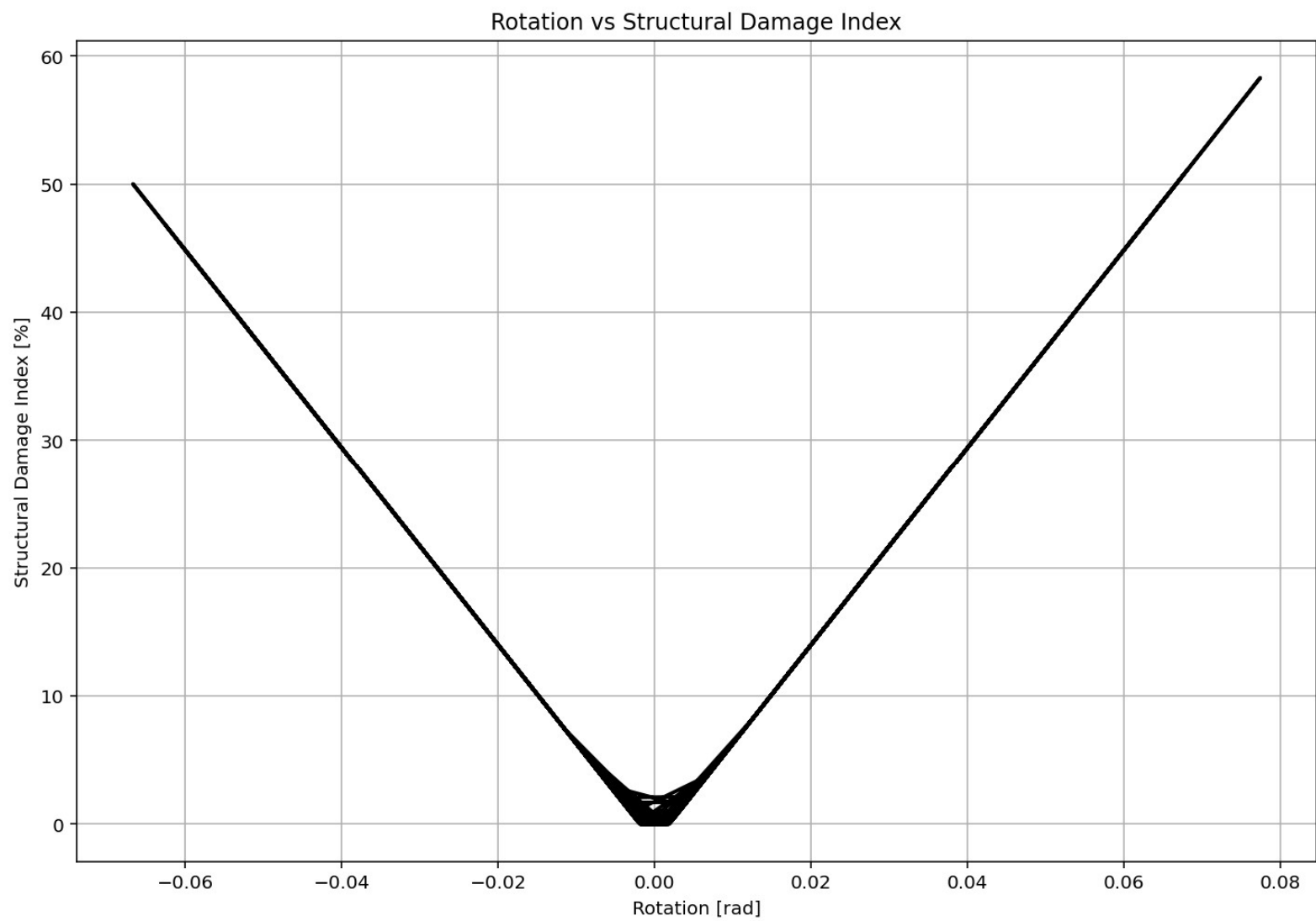
Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 614, Col 5 UTF-8 CRLF RW Mem 48%



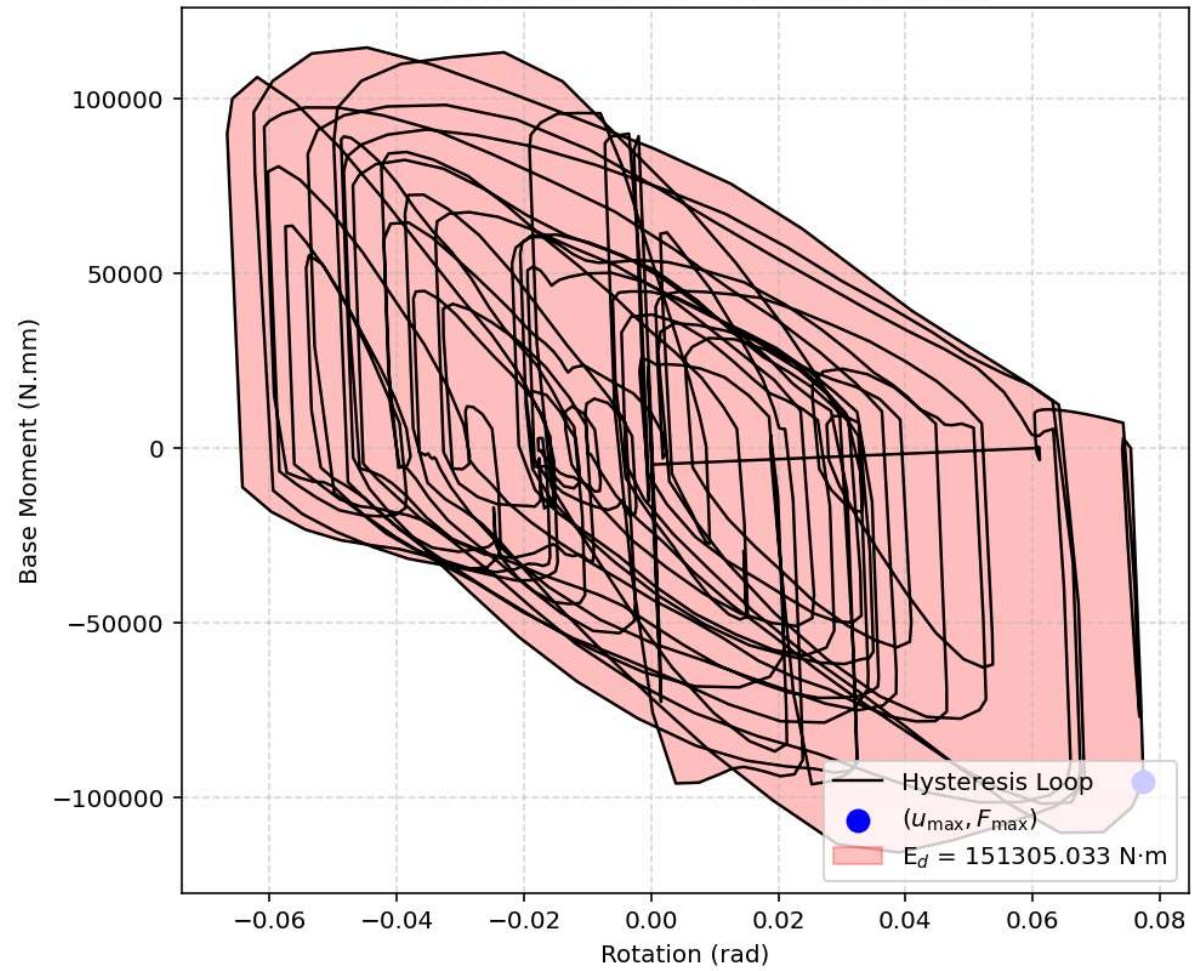


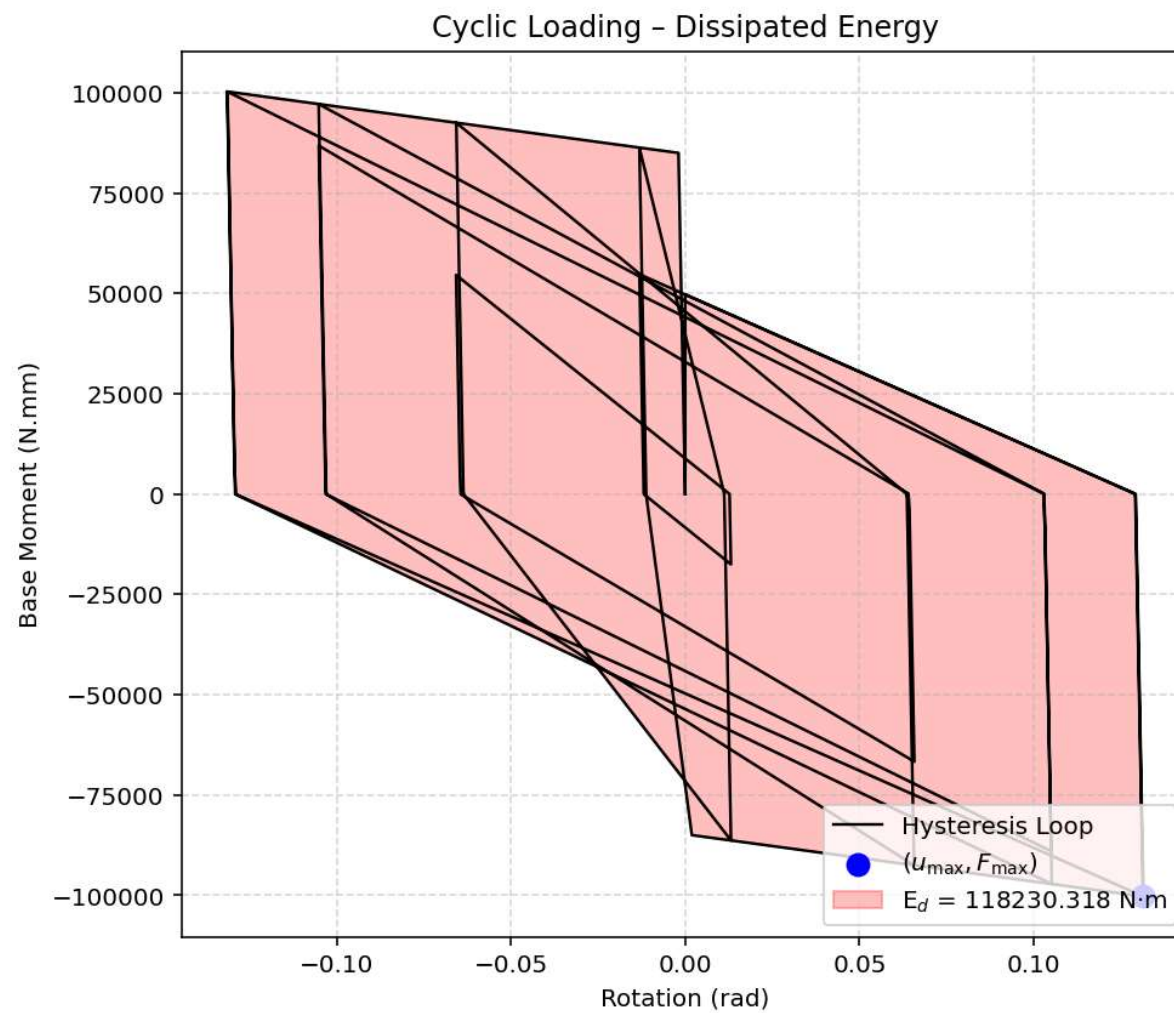
Period of Structure During Seismic Analysis





Earthquake Response - Dissipated Energy





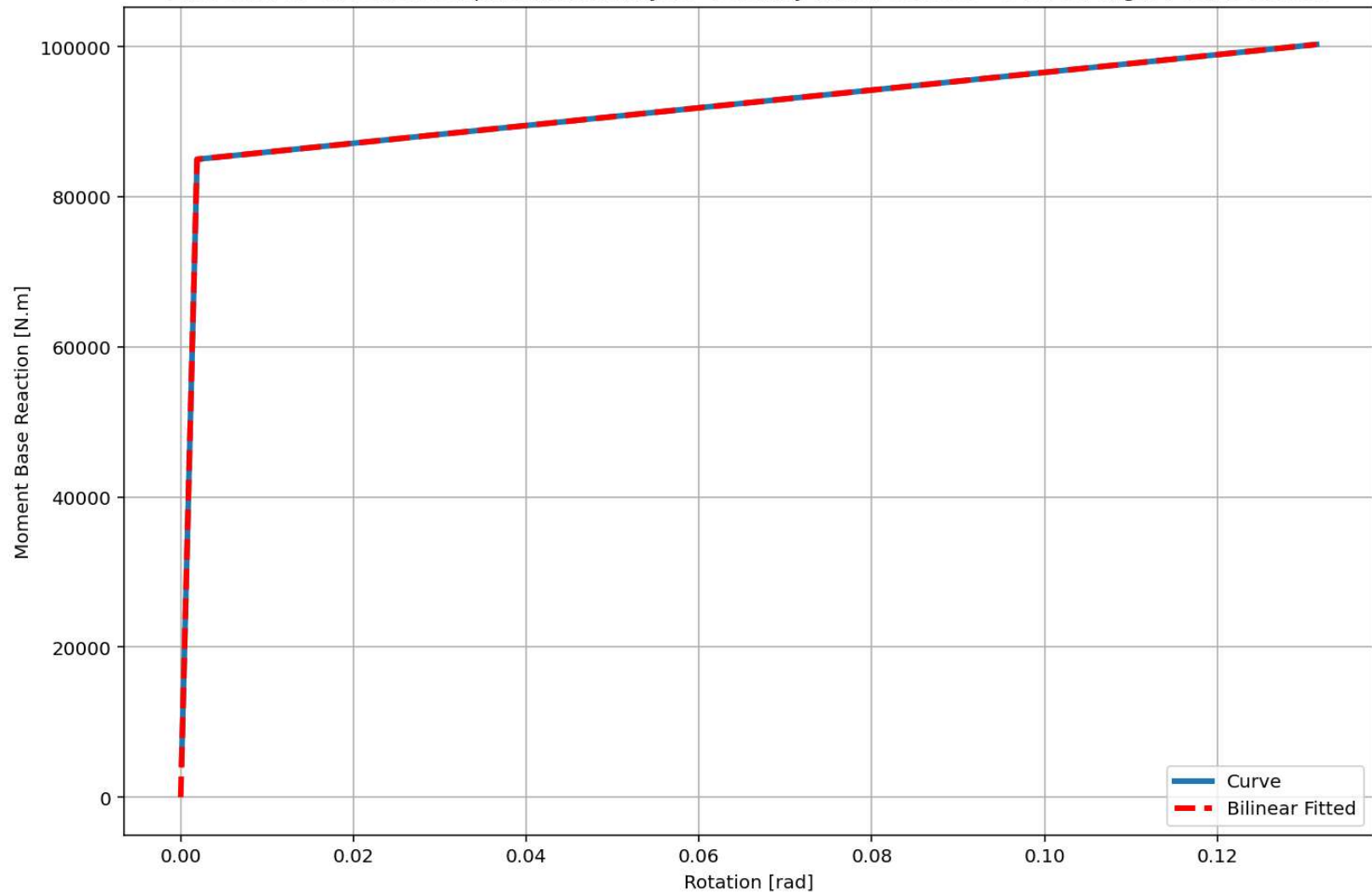
Dissipated Energy from Earthquake= 151305.03 N·m.Rad

Dissipated Energy from Cyclic Rotation= 118230.32 N·m.Rad

FOR EVALUATION OF DISSIPATED ENERGY CAPACITY INDEX:

**CHECK THE CYCLIC ROTATION ANALYSIS AND IF IT IS POSSIBLE
INCREASE THE ROTATION.**

Last Data of BaseShear-Displacement Analysis - Ductility Ratio: 69.6208 - Over Strength Factor: 1.1801



+=====+

= Analysis curve fitted =

Rotation Base Moment

[[0.00000000e+00 0.00000000e+00]

[1.88880391e-03 8.49961760e+04]

[1.31500000e-01 1.00300000e+05]]

+=====+

+-----+

Structure Elastic Stiffness : 45000000.00

Structure Plastic Stiffness : 762737.64

Structure Tangent Stiffness : 118074.86

Structure Ductility Ratio : 69.62

Structure Over Strength Factor: 1.18

+-----+

Over Strength Coefficient (Ω_0): 1.1801

Rotation Ductility Ratio (μ): 69.6208

Ductility Coefficient (R_μ): 69.6208

Structural Behavior Coefficient (R): 82.1562

Structural Ductility Damage Index in X Direction: 58.2884 (%)

Seismic Fragility Curves by Damage State from DYN. TIME-HISTORY ANA. DUCTILITY DAMAGE INDEX [%]

